

SAP BTP Documentation - DEVELOPMENT

BTP Documentation: development

URL: <https://help.sap.com/docs/btp>

Title: SAP Business Technology Platform (SAP BTP) | SAP Help Portal | SAP Help Portal

Home

SAP Business Technology Platform (SAP BTP)English

Extend and personalize SAP applications or integrate and connect entire landscapes with SAP BTP. Unleash your business users to connect processes and experiences, decide confidently with integrity and continuous business innovation.

Provide feedback on our search

à

àFavorite

àDownload PDF

á=Share

SAP BTP Fundamentalsà^

Explore foundational resources for SAP BTP that help you manage the platform effectively.

Getting Started

Gain insights into the essentials and first steps of using SAP BTP.

Basic Platform Concepts

Understand the concepts relevant for managing the platform.

Manage and Operate Accounts and Applications

Learn about administering accounts and applications effectively.

Development

Discover how to create and implement projects using the SAP BTP runtimes.

Security

Inform yourself about ensuring data protection and safety on SAP BTP.

Feature Scope Description for SAP BTP, Cloud Foundry, ABAP, and Kyma Environmentsà

Get an overview of the features available in the Cloud Foundry, ABAP, and Kyma environments.

SAP BTP Adoption Framework(Information published on SAP site)

Explore the SAP BTP Adoption Framework, which provides a central access point for architects, developers and administrators to build and run enterprise-grade solutions on SAP BTP.

Stay Currentà^

Discover the latest features and updates for SAP BTP.

What's New for SAP BTP

Stay updated on the latest features and updates.

SAP Community(Information published on SAP site)

Engage with the SAP community for inspiration, collaboration, and product news.

Release Navigator(Information published on SAP site)

Find consolidated release resources and schedules for SAP BTP products and services.

What's New for SAP BTP (SAP Sovereign Cloud Offerings)

Stay updated on the latest features and updates for sovereign cloud offerings.

Recommended for Administratorsà^

Get advice and tips on optimizing SAP BTP usage and administration within your organization. For service-specific administration, access the service documentation directly.

SAP BTP Administrator's Guide

Plan your account and security model, and set up your landscape and lifecycle management for running applications on SAP BTP.

SAP BTP Security Recommendations

Adopt recommended security measures when architecting solutions.

SAP BTP Multi-Region Guide

Optimize your BTP architecture with effective strategies to set up a multi-region framework, ensuring seamless failover and minimizing downtime for mission-critical applications.

Recommended for Developersà^

Access step-by-step tutorials and documentation crafted specifically for developers working with SAP BTP. For service-specific development, access the service documentation directly.

SAP BTP Developer's Guide

Explore SAP BTP Developer's Guide to help you in developing, delivering, and integrating business applications.

Preparing and Testing Your Solution for Third-Party Cookie Deprecation

Get your SAP BTP solution tested for the upcoming Google Chrome third-party cookie deprecation.

Getting and Applying a Deprecation Trial Token for the Third-Party Cookie Deprecation

Learn about the deprecation trial for the upcoming Google Chrome third-party cookie deprecation and how to get and apply a deprecation trial token for your solution.

Tutorials(Information published on SAP site)

Explore step-by-step tutorials and get started quickly.

APIs, Business Processes, Integrations(Information published on SAP site)

Access the APIs, business processes, and integrations available for SAP BTP.

DevOps with SAP BTP

Explore how DevOps practices can enhance your workflow.

Recommended for Solution Designers^

Find architecture tips and real-world SAP BTP use cases that help you design efficient solutions.

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Learn about the Kyma and Cloud Foundry runtimes, their similarities, synergies, and differences.

Discovery Center Missions(Information published on SAP site)

Implement use cases with step-by-step guidance.

Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP (Cloud Foundry and Kyma)

Learn how to migrate your scenarios from the Neo environment to the multi-cloud foundation.

Application Development and Automation^

Create apps faster, automate processes, and scale without limits.

ABAP Environment

Business Application Studio

Cloud Logging

Cloud Portal Service

Document Management Service

Forms Service by Adobe

Intelligent Situation Automation

Market Rates, Bring Your Own Rates

Market Rates, Refinitiv

Mobile Services

View all

Integration^

Connect and automate your processes.

Advanced Event Mesh

Cloud Integration Automation Service

Connectivity

Event Mesh

Integration Suite

Cloud Integration for Data Services

SAP Cloud Application Event Hub

Master Data Integration

Data and Analytics^

Maximize the value of SAP data and deliver trusted insights.

Data Quality Services

Master Data Governance, Cloud Edition

Object Store

PostgreSQL

Redis

SAP Analytics Cloud
SAP ASE Service
SAP Data Intelligence Cloud
SAP Datasphere
SAP HANA Cloud
âView all
Foundationâ
Operate and secure your applications.
Agent Activation for Dynatrace
Application Autoscaler
Application Logging Service
Audit Log Service
Authorization and Trust Management Service
Automation Pilot
Cloud Foundry Runtime
Cloud Identity Services
Cloud Transport Management
Content Agent Service
âView all
Artificial Intelligenceâ
Infuse artificial intelligence in your apps in a scalable and responsible fashion.
Data Attribute Recommendation
SAP Document AI
Personalized Recommendation
SAP AI Core
SAP AI Launchpad
SAP Translation Hub

URL: <https://help.sap.com/docs/btp/best-practices/best-practices-for-sap-btp>

Title: SAP BTP Administrator's Guide | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's Guide

SAP BTP Administrator's GuideEnglish

Provide feedback on our search

à

SAP BTP Administrator's Guide

This guide is the starting point for setting up application lifecycle management for your specific use case, business, and IT landscape. It contains recommendations and best practices that give you an overview of what you should consider when planning development projects on SAP BTP – from setting up the correct organizational structure to creating an account and security model, to developing and operating applications. It also includes links to step-by-step instructions when required.

Is This Guide for You?â

If you're an architect or a development project lead, this guide helps you plan the architecture of your application and establish boundary conditions to foster the required level of governance and compliance for your development teams (such as by defining the best-fitting delivery and observability concepts).

If you're an administrator, this guide helps you define the correct organizational structure and create an account and security model. Also, it enables you to understand and plan the tasks required to run your organization's applications on SAP BTP reliably - and which SAP BTP services to consider for doing so.

If you're an SAP partner, this guide helps you to set up SAP BTP for running production services for your customers.

Tip

If you're a developer, you have to define the correct methodologies and tools for your development project. In this case, use the SAP BTP Developer's Guide.

Note

This guide is targeted at customers who want to run and use applications in a production environment. If you're an SAP BTP trial user, you might still find that some information in this guide is useful. Check out the following

page for more details about trial accounts: [Trial Accounts and Free Tier](#). Please note that the services available in the trial version differ from the ones in the enterprise version.

Prerequisites

If you're new to SAP BTP, read:

[SAP Business Technology Platform](#)

SAP BTP is a cloud platform that brings together application development, automation, data management, analytics and planning, integration, and AI capabilities into one unified environment optimized for SAP applications. It offers:

An intuitive, modern development environment for both, professional IT as well as citizen and business developers

Built-in data models, integrations, workflows, app templates, and AI business services

Self-service data discovery, modeling, planning, and analytics for business users in a governed environment

[SAP BTP Adoption Framework](#)

This is the central access point for architects, developers, and administrators to build and run enterprise-grade solutions on SAP BTP. It comprises decision guides, reference architectures, methodologies, recommendations, and DevOps principles.

[Basic Platform Concepts](#)

Contains regions, runtimes, accounts, members, quotas, capabilities and links to in-depth explanations.

[Getting Started Checklist](#)

Explains your responsibilities and SAP's responsibilities when it comes to application lifecycle management.

[SAP BTP Developer's Guide](#)

This guide helps you improve the process of implementing a business application on SAP BTP. It's part of the SAP BTP Guidance Framework and contains recommendations and best practices that give you an overview of what you should consider when working on development projects on SAP BTP.

[Administering SAP Business Technology Platform](#)

This learning journey provides an extensive, well-structured collection of links to resources such as videos, blog posts, openSAP courses, and additional documentation.

Check out the [SAP Discovery Center](#) to get an overview on the available SAP BTP services, discover missions that guide you step-by-step, and explore reference architectures.

[How to Use This Guide](#)

Plan and set up your landscape to manage the lifecycle of your cloud applications.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.

Set Up and Plan – build teams, set up your account and security model, and create an enrollment process for your development projects.

Develop and Build – find out about the tools and programming languages that are available on SAP BTP, and how to use multitarget applications to efficiently manage dependencies.

Deploy and Deliver – deploy and deliver simple and multitarget applications.

Integrate and Test – test and integrate your application with other solutions.

Go Live and Operate – learn what's important for going live and operating applications, services, and hybrid landscapes.

Improve and Retire – make improvements to your application, perform housekeeping, and learn about what's important to consider when you want to retire it.

[Contribute to This Guide](#)

Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

Feedback Create issue: Provide feedback about a documentation page. This option opens an issue on GitHub.

Feedback Edit page: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

[More information:](#)

[Contribution Guidelines](#)

[Introduction Video](#)

URL: <https://help.sap.com/docs/btp/btp-admin-guide/0f79436cf43a4785bb5b7ee9dac0c76a.html>

Title: Neo Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsEnvironmentsNeo Environment

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

See also:

What Is SAP BTP, Neo Environment

Development, Neo Environment

Extensions, Neo Environment

Administration and Operations, Neo Environment

Security, Neo Environment

Getting Support, Neo Environment

Migration from the Neo to the Multi-Cloud Foundation^

To learn more about why and how to migrate your scenarios on SAP BTP, see [Migrating from the SAP BTP Neo Environment to the Multi-Cloud Foundation](#).

URL: <https://help.sap.com/docs/btp/btp-admin-guide/deploy-and-deliver>

Title: Deploy and Deliver | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's GuideDeploy and Deliver

SAP BTP Administrator's GuideEnglish

Provide feedback on our search

à

Deploy and Deliver

After you've developed your application, you need to deploy it and deliver it to your TEST and PROD environments. Furthermore, consider implementing failover.

If you have only a simple application (such as for a small proof of concept) and don't have the need for governance and scalability, you can consider performing a manual deployment, to bring your application into the corresponding runtime in your landscape. For deploying an application manually, you can choose between several native options, depending on your environment and application.

In enterprise environments, a managed and automated delivery approach is recommended – because it is less error-prone and creates repeatable outcomes. Furthermore, you can apply governance and central control of the propagation of your changes towards your production environment. For delivery, you can choose between several options, depending on your requirements and your environment.

Also, don't forget to implement failover, for example, by deploying your application to two different datacenters in parallel.

Related Information^

Deploying Applications

Delivering Applications

Implementing Failover

URL: <https://help.sap.com/docs/btp/btp-admin-guide/develop-build>

Title: Develop and Build | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's GuideDevelop and Build

SAP BTP Administrator's GuideEnglish

Provide feedback on our search

à

Develop and Build

The whole process and details for developing an application are described in the [SAP BTP Developer's Guide](#).

Based on the experiences of successful business applications, that guide condenses best practices and technologies that can be safely recommended to you. These recommendations are structured following the design-led development phases:

Explore

Identify the business opportunity and set up the roles and responsibilities in your team.

Discover

Identifying Required Use Cases

Understand your end users, their working methods, and their needs in more depth.

Understanding Available Technology

Development Use Cases

Get to know the use case patterns that can be implemented on SAP BTP.

Business Application Services

Explore the comprehensive set of tools and services that allow you to keep the pace when developing your application and at the same time benefit from future innovations in SAP BTP.

Design

User Experience Design

Design the user experience of your application to provide good usability.

Technology Design

Use domain-driven design to create an application that is modularized and has shorter innovation cycles and a clearer focus.

Deliver

Set Up

Before you begin developing your applications, make sure your landscape setup is appropriate for managing their lifecycles.

Develop

Develop your application by following the recommendations outlined in this section about the two programming models, the numerous programming languages, the respective development tools and also about aspects such as multitenancy, security, extensibility and others.

Run and Scale

Get constant feedback from your users, optimize the application based on this feedback, and operate the application using the services and capabilities SAP BTP offers.

Programming Model Specifics

ABAP Cloud

ABAP Cloud reflects the modern way to develop ABAP. It allows you to build lifecycle-stable and cloud-ready business applications, services, and extensions.

SAP Cloud Application Programming Model (CAP)

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It supports Java (with Spring Boot), JavaScript, and TypeScript (with Node.js), which are some of the most widely adopted languages. CAP guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

URL: <https://help.sap.com/docs/btp/btp-admin-guide/set-up-and-plan>

Title: Set Up and Plan | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's GuideSet Up and Plan

SAP BTP Administrator's GuideEnglish

Provide feedback on our search

à

Set Up and Plan

Before you begin developing your applications, make sure your organizational and landscape setup is appropriate for managing their lifecycles, and consider failover to prevent breakdowns.

Creating a Governance Model

One of the first and most important steps of your journey to the cloud is to establish an appropriate organizational setup and corresponding governance model. A clear and well-thought-out organizational setup makes it easier for your employees to adopt agile processes.

Setting Up Your Account Model

Learn how to set up your account model with global accounts and subaccounts, and how to use directories, spaces and namespaces to match your business and development needs.

Setting Up Your Security and Compliance Model

Applications on SAP BTP are exposed to the Internet and should therefore fulfill the highest possible security requirements to prevent unauthorized access.

Planning Failover on SAP BTP

Use a multi-data center setup and implement automatic failover to ensure the high availability of your applications in case of a data center outage.

Managing Cost

Consumption-based commercial models offer flexibility to optimize your costs. To manage these costs and avoid charges for overage, it's recommended that you monitor your usage and consumption.

URL: <https://help.sap.com/docs/btp/btp-developers-guide-dev/btp-developers-guide>

Title: *Page Not Found | SAP Help Portal*

ProductsWhat's NewExplore SAP

Sorry

We couldn't find that page

Go back to the SAP Help Portal home page.

Copyright

Disclaimer

Privacy Statement

Legal Disclosure

Trademark

Terms of Use

Cookie Preferences

Accessibility & Sustainability

Ask a Question about the SAP Help Portal

Find us on

Share

á=

Your choices regarding cookies on this site

This site uses cookies and related technologies, as described in our Cookie Statement, for purposes that may include site operation, analytics, enhanced user experience, or advertising. You may choose to consent to our use of these technologies, or manage your own preferences.

[Privacy Statement](#) | [Cookie Statement](#)

[Accept All](#)

[Deny All](#)

[Manage Settings](#)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/00aa2c23479d42568b18882b1ca90d79.html>

Title: *Entitlements and Quotas | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsEntitlements and Quotas

SAP Business Technology PlatformEnglish

[Provide feedback on our search](#)

à

Entitlements and Quotas

When you purchase an enterprise account, you're entitled to use a specific set of resources, such as the amount of memory that can be allocated to your applications.

Services and Service Plansà^

On SAP BTP, all external dependencies such as databases, messaging systems, files systems, and so on, are services. In this context, multitenant applications and environments are considered services.

Each service has one or more service plans available. A service plan is the representation of the costs and benefits for a given variant of a particular service. For instance, a database may be configured with various "T-shirt sizes", each of which is a different service plan.

Entitlements

An entitlement is your right to provision and consume a resource. In other words, entitlements are the service plans that you're entitled to use.

The entitlements that are available in your enterprise account are determined by the commercial model used by your global account:

In the consumption-based commercial model, your organization buys an entitlement to all current and future SAP BTP services eligible under this model. During your contract period, you have full flexibility to switch services on and off and change between them as your business needs. For more information, see [What Is the Consumption-Based Commercial Model?](#).

In the subscription-based commercial model, your organization is authorized to use only the services you have subscribed to, as per your commercial agreement. To access additional services at an extra cost, you can modify your contract through your sales representative or account executive. For more information, see [What Is the Subscription-Based Commercial Model?](#).

Quotas

A quota represents the numeric quantity of a service plan that you're entitled to consume in your global account and its subaccounts.

From the perspective of your SAP BTP global account contract, the global quota of a service plan can be defined as either fixed or unlimited:

Fixed quota: The quota or allowance of the service plan is the upper limit at which the plan can be consumed collectively across your global account. This is typical of global accounts that have the subscription-based commercial model agreement.

Unlimited quota: There is no upper limit to how much the service plan can be used collectively across your global account. This is typical of most services that are eligible to global accounts with a consumption-based commercial model agreement (Cloud Platform Enterprise Agreement (CPEA) and Pay-As-You-Go for SAP BTP). At the end of the month, the number of units used determines how much you are charged.

Regardless of whether the global quota of a plan on the global account level is fixed or unlimited, the specifics and properties of each plan determines how its quota can be distributed by the global account admin to the subaccounts in your global account. There are two types of quota assignments:

Numeric assignments: The global account admin assigns the plan to a subaccount and specifies the maximum quantity of units from the global quota that can be consumed by the subaccount. Admins can use this assignment to limit users from exceeding a specific quota for cost-controlling purposes. Typically, but not always, the assigned quota is directly related to the number of instances that can be created in subaccounts for the assigned plan.

Non-numeric assignments: The global account admin does not assign a specific quantity but simply grants access to the specific plan by assigning the plan to the subaccount. When the plan is assigned, it's available for use by the subaccount. SaaS application assignments are typically non-numeric, whereby a subaccount can either subscribe to the app or not. For services that fall into this category of non-numeric assignments, there is usually no limit to the number of instances that can be created.

See also the sections below and [Managing Entitlements and Quotas Using the Cockpit](#).

Distribution and Usage of Entitlements and Quotas

Entitlements and quotas are purchased and managed at global account level, from where they're distributed to subaccounts, which consume them.

Tip

You can also distribute entitlements to directories, and then redistribute to the subaccounts under the directories; however, this is optional. By default, directories are not enabled to manage entitlements. This feature needs to be enabled per directory by the global account administrator.

When assigning entitlements and quotas to directories, you also have the option to automatically assign a set amount of quota to each new subaccount added to the directory. This option doesn't apply to subaccounts that are already in the directory when you select this option. The quota you assigned to the directory is then gradually distributed to all subaccounts that you add to that directory, until it runs out. Once the directory quota runs out, if you add a new subaccount to that directory it won't get any quota automatically anymore.

Since directories are only a way of grouping subaccounts, you can't consume a service at directory level.

However, when you assign entitlements and quotas from the global account to a directory, the quota you assigned is shown as used, even if there are no subaccounts in that directory to consume the quota. You can

think of it as a way to "reserve" quota and make sure it's not assigned to other subaccounts or directories. When you remove quotas or assignments from a directory or subaccount, they become available again at global account level and can be assigned to other directories or subaccounts; unless the quota is reserved for a given directory then the freed quota remains available only to that directory and its subaccounts.

Note

Before a subaccount admin can enable a quota-based environment, such as Kyma, the subaccount admin must first assign the environment as an entitlement to the subaccount. Other environments, such as Cloud Foundry, are available by default to all subaccounts, and therefore are not available as entitlements.

For more information, see:

[Configure Entitlements and Quotas for Subaccounts](#)

[Configure Entitlements and Quotas for Directories](#)

[Quota Plans](#)

In the Cloud Foundry environment, you can further distribute the quotas that are allocated to a subaccount.

This is done by creating space quota plans and assigning them to the spaces.

Space quota plans are optional and are used to limit how much each space can use, not to enable. If you don't create any space quota plans, all spaces in a subaccount using the Cloud Foundry environment have access to all entitlements and quotas in that subaccount. This means that one space could use up all the quota in a subaccount. If you want to prevent that from happening and set a limit to how much each space can use from the total quota available in the subaccount, you can use quota plans and assign them to spaces.

For more information on space quota plans in the Cloud Foundry environment, see:

<https://docs.cloudfoundry.org/adminguide/quota-plans.html>

[Managing Space Quotas](#)

[Resource Providers](#)

SAP BTP allows you to connect your global account in the SAP BTP Cockpit to your provider account from a non-SAP cloud vendor, and consume remote service resources that you already own and which are supported by SAP through this channel.

For example, if you're subscribed to Amazon Web Services (AWS) and have already purchased services, such as PostgreSQL, you can register the vendor as a resource provider in SAP BTP and consume this service across your subaccounts together with other services offered by SAP.

SAP BTP currently supports the following vendors and their consumable services:

[Cloud Vendor](#)

[Supported Services](#)

[Amazon Web Services](#)

[Amazon Relational Database Service \(RDS\) - PostgreSQL](#)

[Microsoft Azure](#)

[Azure Database for PostgreSQL](#)

After you configure a new resource provider, its supported services are added as entitlements in your global account. In the Entitlements page in the cockpit, you can then allocate the required services and quotas to the relevant directories and subaccounts in your global account.

For more information, see [Managing Resource Providers](#).

[Related Information](#)

[Managing Entitlements and Quotas Using the Cockpit](#)

[Tutorial: Manage Entitlements on SAP BTP Trial](#)

[Troubleshooting: Entitlement and Quota Problems](#)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/046f127f2a614438b616ccfc575fdb16.html>

Title: [Trial Accounts and Free Tier | SAP Help Portal](#)

Home [SAP Business Technology Platform \(SAP BTP\)](#) [SAP Business Technology Platform](#) [Basic Platform](#)

[Concepts](#) [Trial Accounts and Free Tier](#)

[SAP Business Technology Platform](#) [English](#)

[Provide feedback on our search](#)

[à](#)

[Trial Accounts and Free Tier](#)

[Explore the different options for trying out SAP BTP.](#)

[Trial Account or Free Tier Offering: When to Use Which?](#)[à](#)

Before setting up your account, you need to decide which free offering for SAP BTP is suitable for your needs: Pay-As-You-Go or CPEA accounts with free tier service plans are open to customers, partners, and let you try out SAP BTP in a per service defined time span for free. These account types enable you to test your scenarios and generally offer the option to upgrade to paid service plans. These accounts also allow you to store data long-term and move projects to production. You also get access to our community, including free technical resources such as tutorials and blog posts. For more information, see [Enterprise Accounts and Using Free Service Plans](#).

Note

When using Kyma, the free plan is available only once in a global account for up to 30 days. For more information, see [Available Plans in the Kyma Environment](#).

You can self-register for an enterprise account with free tier service plans. For more information, see [Get an Account on SAP BTP to Try Out Free Tier Service Plans](#).

Note

Only community support is available for free tier service plans and these are not subject to SLAs.

Note

The option to upgrade from free tier service plans to paid service plans is not yet available for all services and runtimes, such as Kyma. See [Using Free Service Plans](#) to read more about upgrading from free tier service plans to paid service plans.

A trial account lets you try out SAP BTP for free for 90 days. The services provided for the trial account allow restricted use of the platform resources and services. Access is open to everyone. Trial accounts are intended for personal exploration, your own non-productive testing, and evaluation of the services in accordance with SAP BTP trial terms and conditions. A SAP BTP trial account must not be used for production use or team development. You are not permitted to use the trial account in any productive or commercial manner.

You can self-register for a trial account. For more information on how to do that, see [Get a Free Account on SAP BTP Trial](#)

[Free Tier and Always Free Tags in the SAP Discovery Center](#)

In the SAP Discovery Center service catalog overview, some services display the tag Free Tier, which indicates the service is offering a free tier service plan. Some services display the tag Always Free, which indicates the service is offering a service plan that comes free of additional charges, as it is already included in your overall SAP BTP contract. You can use the filter function of the SAP Discovery Center to filter for services that offer Free Tier and Always Free plans. The Always Free service plans include the following note in their service plan description: "This service plan is included in the overall SAP Business Technology Platform contract".

Note

Always Free service plans might not be available in all regions or for all providers.

[Trial Lifecycle](#)

Use your SAP BTP trial account:

Familiarize yourself with the Trial Scope or try out one of our Starter Scenarios on the Tutorial Navigator.

Extend your SAP BTP trial account in intervals:

Your 90-day trial period is divided in intervals. If you sign in to your trial account regularly, the intervals are extended automatically for you by up to 30 days or until your overall 90-day trial is finished.

If you don't sign in to your trial account for 30 days or more, your account will be suspended. All suspended accounts will be deleted and you won't be able to use applications or services.

Delete your SAP BTP trial account:

After 90 days, your trial account is automatically deleted. But if you want to proactively delete your SAP BTP trial account, you can navigate to the global accounts scope and select the Account Explorer page, then click the Delete Trial Account button.

If you want to continue to use an SAP BTP trial account after deletion, you need to set up a new account.

If you want to explore SAP BTP without time limit, create an enterprise account with free tier service plans allowing you to test your scenarios. See: [Get an Account on SAP BTP to Try Out Free Tier Service Plans](#).

Caution

Environments enabled on your trial account may not be valid for your whole trial period. To learn about the validity period for the Kyma environment, see [About the Trial Account](#).

[Trial Scope](#)

A trial account enables you to explore the basic functionality of SAP BTP.

SAP BTP trial accounts are available in several regions. For more information, see [Regions](#).

You can create directories in your trial account. For more information, see [Managing Directories Using the Cockpit](#)

You can use productive and beta services. To consume beta services, you must enable the subaccount for beta features during the subaccount creation or when you edit the subaccount details.

You can manage platform users by assigning them role collections. For more information, see [Working with Role Collections](#).

A trial account includes 4 GB of memory for applications.

You can use 8 GB of instance memory.

There are 10 total routes and 40 total services available.

You can use 2 configured on-premise systems with the Cloud connector.

There's no service level agreement with regards to the availability of the platform.

You can use HDI containers in a shared SAP HANA database (only available on cf-us10).

For cleanup purposes, applications stop automatically on a daily basis. You need to manually restart them when needed.

Note

Applications are stopped at midnight (or some time later depending on server load) relative to the region in which you created your trial account. If you're working in a time zone that is far from the region where your trial account was created, then your applications may stop during business hours.

You can have a trial tenant of SAP Cloud identity Services, see [Get Your Tenant](#).

Related Information

[Getting Started with a Trial Account in the Cloud Foundry Environment](#)

[Getting Started with a Trial Account in the Kyma Environment](#)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/apis>

Title: [APIs | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...Programming Languages, Tools, and APIsAPIs

SAP BTP Developer's GuideEnglish

[Provide feedback on our search](#)

à

APIs

Discover and consume APIs to manage, build, and extend the core capabilities of SAP BTP.

An Application Programming Interface or API is an interface provided by an application for interacting with other applications. APIs specify how software programs are able to exchange information with each other, even if designed and run by different organizations. They facilitate interaction by selectively exposing certain functionality, allowing different apps, websites, and devices to communicate effectively with each other. More importantly, APIs enable businesses to reach beyond regular business channels and share data, content, and services directly to both B2B (business to business) and B2C (business to consumer) clients, making UI development easy.

API Offerings on SAP BTP

SAP BTP enables you to consume APIs and publish your own ones through the following offerings:

Offering

Description

[SAP BTP on SAP Business Accelerator Hub](#)

The SAP Business Accelerator Hub provides you with one central repository for browsing and accessing APIs from SAP and selected partners. Test APIs and try out mock data in your systems.

It is also the official place where REST and OData REST API references are published.

[API Management Capability within SAP Integration Suite](#)

With the API Management capability, you can build, manage, publish, and monetize your own APIs within one secure and scalable environment.

[Software Development Kits \(SDKs\)](#)

For an overview of the available SDKs, see [Tools](#).

The SDKs available for SAP BTP offer APIs to, for example, accelerate enterprise app development.

Related Information

URL: <https://help.sap.com/docs/btp/btp-developers-guide/b39bae31d35d4d039431973116363d57.html>

Title: *Delivering Applications | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's GuideDeploy and

DeliverDelivering Applications

SAP BTP Administrator's GuideEnglish

Provide feedback on our search

à

Delivering Applications

In enterprise environments, use a managed and automated delivery approach – because it is less error-prone and creates repeatable outcomes. Furthermore, you can apply governance and central control of the propagation of your changes towards your production environment.

Overviewà^

You have several options for propagating developed applications and application-specific content from one subaccount to another automatically and with more control; for example, from your Development or Testing to your Production subaccounts.

The delivery approaches are based on some of the concepts and approaches as outlined in Deploying Applications.

The following table presents the delivery options, with tool support and automation decreasing from top to bottom:

Delivery Option

Development Content (such as Java, HTML5, CAP applications)

SAP Cloud Integration Content

SAP HANA Cloud Artifacts

SAP Build Work Zone

Other App-Specific Content

Kyma-Based Apps

SAP Continuous Integration and Delivery (CI/CD)

recommended

Basic pipeline available, still in development

recommended

n/a

n/a

recommended (automating the packaging into a Dockerfile, and the deployment)

SAP Cloud Transport Management with the option to integrate into Change Management*

recommended (as MTA)

recommended (as MTA)

recommended (as MTA)

recommended

recommended

n/a

CTS+ with the option to integrate into Change Management*

recommended (as MTA)

recommended (as MTA)

recommended (as MTA)

n/a

n/a

n/a

Manually

For more information, see Deploying Applications.

àf

* SAP Cloud ALM change and deployment management, and SAP Solution Manager Change Request Management and Quality Gate Management

Delivery Optionsà^

Recommendation

For Cloud Foundry, use continuous integration together with transport management, to combine agility with control:

Your development team can benefit from the highly automated CI process that automatically qualifies their changes after submitting them to the source code repository. Also, this approach makes sure that only qualified changes are considered for propagation towards your production environment. Especially with SAP Continuous Integration and Delivery, you can drastically reduce the cognitive load of your development teams.

Still, you can benefit from additional control for the propagation of your changes towards your production environment, where you centrally define delivery landscapes and configure who is allowed to handle changes where, which is required in most enterprise environments and regulated markets. Also, you get the option to integrate into new or existing change management processes, such as offered by SAP Cloud ALM – to synchronize transport of changes in hybrid scenarios.

If you're not facing the (immediate) need for more control, SAP recommends starting with SAP Continuous Integration and Delivery, if it covers your development scenario. You can easily activate the automated hand-over into transport management at a later point in time. For example, you could perform direct deployment from the pipeline during a proof of concept, but then integrate into transport management when the team comes up with a release candidate.

For Kyma, use a CI/CD pipeline to package your application into a Docker image and the corresponding deployment. For this, create Helm charts that you can link to your pipeline. This way, you can benefit from automation and a higher reliability. Start with the corresponding pipeline template of SAP Continuous Integration and Delivery. Only if you need more sophisticated pipelines and delivery landscapes, consider building your own pipeline with Project "Piper".

To learn more about the delivery options, expand the sections:

SAP Continuous Integration and Delivery (CI/CD)

Transport Management

CI/CD with Transport Management

Manually

Related Information

Blog Post: Transport Approaches in SAP BTP

Blog Post: Getting started with the predefined CI/CD pipeline for Kyma runtime

Blog Post: Cloud Native Buildpack Support in SAP Continuous Integration and Delivery

Example: Deliver a Multi-Target Application Using Continuous Integration and Transport Management

When developing applications for SAP BTP, you can combine the agility of continuous integration with the control of transport management.

You create the application with the IDE of your choice, for example SAP Business Application Studio. The artifacts are stored in a central source code repository.

Whenever a developer commits a change to the source code repository, a continuous integration pipeline starts.

Recommendation

For typical SAP BTP scenarios, such as development with SAP Cloud Application Programming Model, we recommend using SAP Continuous Integration and Delivery. The pipeline templates contain steps to build the application, run automated tests, and deploy the application to your development subaccount.

In the "release" stage of the pipeline, you can also hand over the MTA to SAP Cloud Transport Management. This creates a transport request, which can then be handled according to your governance requirements.

Recommendation

We recommend this step only for qualified release candidates of your application. This can be achieved by using a separate release pipeline, for example.

Optionally, you can integrate SAP Cloud Transport Management with SAP Cloud ALM or SAP Solution Manager. With that, you can manage hybrid projects, including SAP BTP and on-premise parts.

Related Information

Continuous Integration and Delivery Best Practices Guide

Transporting Multitarget Applications with CTS+

URL: <https://help.sap.com/docs/btp/btp-developers-guide/b6a7e11c3a58416a9ab1175bba17193a.html>

Title: Availability Zones in the Cloud Foundry Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Cloud Foundry

Availability Zones in the Cloud Foundry Environment

The Cloud Foundry environment follows the recommendations of our partner IaaS providers by leveraging the availability zones (AZ) concept.

About Availability Zonesà^

Availability zones (AZ) are single failure domains within a single geographical region and are separate physical locations with independent power, network, and cooling. Multiple AZs exist in one region and are connected with each other through a low-latency network.

2-AZ and 3-AZ Deployments

To achieve better fault-tolerance, our partners recommend deploying services across multiple AZs, which improves the availability of a service if there are issues with the region infrastructure of one AZ.

High Availability at Platform and Application Levelà^

The SAP BTP Cloud Foundry environment follows these recommendations to support high availability at the platform and application level:

High availability of the platform components:

The building blocks of Cloud Foundry and the virtual machines on which the Cloud Foundry application instances are scheduled run in a high availability setup. These instances are distributed across different AZs. The technology that manages the deployment of the Cloud Foundry environment monitors the health of the platform. If there are infrastructure failures, it re-creates the faulty components.

High availability on the application level:

We recommend running multiple application instances to increase availability. For more information, see [Run Multiple Instances to Increase Availability](#). On SAP BTP, there are three ways to increase application instances: Scaling your application using the application manifest. The manifest.yml allows you to make and save configurations for your application. To scale, you can configure the instance count in the manifest and push the application again with the new configuration. See [App Manifest Attribute Reference](#). To avoid downtimes when updating your application configuration, you can also consider using rolling application deployments. See [Rolling App Deployments](#).

Scaling your application using the cf scale command in the Cloud Foundry command line interface (CF CLI). See [Scaling an App Using cf scale](#).

Scaling your application using the SAP BTP cockpit. See [Add or Remove Application Instances](#).

The Cloud Foundry container scheduler takes care of distributing the different instances of one application on virtual machines in different AZs. For more information, see [How Diego Balances App Processes](#).

Cloud Foundry is constantly monitoring the health state of application instances and restarts instances that are considered unhealthy. See [Using App Health Checks](#).

When the number of desired instances doesn't match the number of actually running instances, Cloud Foundry reschedules the missing instances, for example, when the virtual machines that an application instance was initially scheduled on become unresponsive.

For more information on high availability configuration, see [High Availability in Cloud Foundry](#).

For more information on application stability and resilience, see [Developing Resilient Applications](#).

URL: <https://help.sap.com/docs/btp/btp-developers-guide/basic-platform-setup>

Title: [Set Up | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...DeliverSet Up

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Set Up

Configuring the landscape setup in its essence is an administrative task described in detail in the SAP BTP Administrator's Guide.

Set Up Your Account Modelà^

The hierarchical structure between global accounts, directories, and subaccounts lets you define an account

model that accurately fits your business and development needs. See [Setting Up Your Account Model](#) in the SAP BTP Administrator's Guide.

[Set Up the SAP BTP Infrastructure Automatically Using the Terraform Provider](#)

The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP. By leveraging this provider, you can simplify and streamline the deployment and maintenance of SAP BTP services and applications. See [Automated SAP BTP Infrastructure Setup](#).

[Manage Users and Members](#)

User accounts enable users to log on to SAP BTP and access subaccounts and use services according to the permissions given to them. In this context, it's important to understand the difference between the two types of users that we refer to: platform users and business users.

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot applications and services on SAP BTP.

Business users use the applications that are deployed in SAP BTP. For example, the end users of SaaS applications or services.

See [User and Member Management](#) in the SAP Business Technology Platform documentation.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/btp-developers-guide>

Title: SAP BTP Developer's Guide | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide

SAP BTP Developer's GuideEnglish

[Provide feedback on our search](#)

à

SAP BTP Developer's Guide

The SAP BTP Developer's Guide is the starting point for developing a business application on SAP BTP. It contains recommendations and best practices that give you an overview of what you should consider when working on development projects on SAP BTP. It also contains links to step-by-step instructions when required.

[Is This Guide for You?](#)

If you're an architect or a development project lead, this guide helps you to plan the architecture of your application and the services on SAP BTP you can use to achieve your goal.

If you're a developer, this guide helps you to define the correct methodologies and tools for your development project.

If you're an SAP partner, this guide helps you to set up SAP BTP for developing and running applications and services for your customers. See [Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors](#).

Note

If you're an administrator, you have to set up the correct organizational structure to create an account and security model. In this case, use the [SAP BTP Administrator's Guide](#).

Note

This guide is targeted at customers who want to run and use applications in a production environment. If you're an SAP BTP trial user, you might still find that some information in this guide is useful. Check out the following page for more details about trial accounts: [Trial Accounts and Free Tier](#). Please note that the services available in the trial version differ from the ones in the enterprise version.

[Prerequisites](#)

If you're new to SAP BTP, read:

[SAP Business Technology Platform](#)

SAP BTP is a cloud platform that brings together application development, automation, data management, analytics and planning, integration, and AI capabilities into one unified environment optimized for SAP applications. It offers:

An intuitive, modern development environment for both, professional IT and citizen and business developers, Built-in data models, integrations, workflows, application templates, and AI business services

Self-service data discovery, modeling, planning, and analytics for business users in a governed environment

[SAP BTP Adoption Framework](#)

This is the central access point for architects, developers, and administrators to build and run enterprise-grade solutions on SAP BTP. It comprises decision guides, reference architectures, methodologies, recommendations, and DevOps principles.

Basic Platform Concepts

Contains regions, runtimes, accounts, members, quotas, capabilities and links to in-depth explanations.

Getting Started Checklist

Explains your responsibilities and SAP's responsibilities when it comes to application lifecycle management.

SAP BTP Administrator's Guide

Helps you plan and set up your landscape and your lifecycle management for running applications on SAP Business Technology Platform (SAP BTP). It's part of the SAP BTP Adoption Framework and contains recommendations for planning development projects – from setting up the correct organizational structure to creating an account and security model, to operating applications.

SAP Discovery Center

Helps you turn your data into business value with SAP BTP. In the SAP Discovery Center you can find end-to-end scenarios, called missions, the Service Catalog of SAP BTP, and reference architectures.

SAP Architecture Center

Provides solution reference architectures, helping businesses adopt SAP solutions to turn data into valuable business insights.

How to Use This Guide

Use the SAP BTP Developer's Guide to help you implement business applications on SAP BTP. This guide is part of the SAP BTP Adoption Framework and explains the building blocks for developing, delivering, and integrating business applications.

Understand the Available Technology

Understanding the development use cases, recommended services, and the continuous integration and continuous delivery (CI/CD) principles offered by SAP Business Technology Platform (BTP) is essential to streamline application development, deployment and integration within the SAP ecosystem.

See Understanding Available Technology.

Select Runtime and Programming Model

Choosing a suitable runtime and the corresponding programming model is crucial to optimize application development, deployment and management on SAP BTP. There are different strengths and benefits you can explore before deciding:

SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP

SAP BTP ABAP Environment with ABAP Cloud

Go Through the Development Phases

Development projects for business applications have similar characteristics. Standardized development guidance is driving developer efficiency.

Based on the experiences of successful business applications, this guide condenses best practices and technologies that can be safely recommended to you. These recommendations are structured following the design-led development phases:

The focus of the SAP BTP Developer's Guide is on the design, deliver and run and scale phases. However, you can also find information related to the explore and discover phases.

Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It supports Java (with Spring Boot), JavaScript, and TypeScript (with Node.js), which are some of the most widely adopted languages. CAP guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

Explore

Identify the business opportunity and set up the roles and responsibilities in your team.

Discover

Design

User Experience Design

Design the user experience of your application to provide good usability.

Technology Design

Use domain-driven design to create an application that is modularized and has shorter innovation cycles and a clearer focus.

Deliver

Set Up

Before you begin developing your applications, make sure your landscape setup is appropriate for managing their lifecycles.

Develop

Develop your application by following the recommendations outlined in this section about the two programming models, the numerous programming languages, the respective development tools and also about aspects such as multitenancy, security, extensibility and others.

Run and Scale

Get constant feedback from your users, optimize the application based on this feedback, and operate the application using the services and capabilities SAP BTP offers.

Developing with ABAP Cloud in the SAP BTP ABAP Environment

ABAP Cloud reflects the modern way to develop ABAP. It allows you to build lifecycle-stable and cloud-ready business applications, services, and extensions.

Contribute to the SAP BTP Developer's Guide[^]

Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

Feedback Create issue: Provide feedback about a documentation page. This option opens an issue on GitHub.

Feedback Edit page: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

Contribution Guidelines

Introduction Video

Introduction Blog Post

URL: <https://help.sap.com/docs/btp/btp-developers-guide/build-and-run-multitenant-saas-application-for-partners-who-are-independent-software-vendors>

Title: Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors | SAP Help Portal HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Build and Run Multitenant SaaS Applications for Partners Who Are ...

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors

If you are an independent software vendor, SAP Business Technology Platform (SAP BTP) offers architecture patterns and development practices for building multitenant SaaS applications, along with operational guidance and tutorials. Utilizing the following resources help you to streamline the development process and ensure scalable and efficient application deployment.

Gain access to a single, comprehensive, and structured resource for partners building multitenant SaaS solutions on SAP BTP. This guide outlines the recommended path, helping you navigate every step - from the initial idea to successful monetization - and makes it easier to fully leverage SAP BTP's capabilities, including its services, scalability, and go-to-market opportunities. See End-to-End Guide for Multitenant SaaS Solutions on SAP BTP on SAP Partner Portal.

Explore the partner reference application. It provides a comprehensive example to help you build, deploy, and run scalable SaaS solutions. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application 'Poetry Slam Manager' on GitHub.

Explore the partner reference application extension that provides tutorials and a comprehensive example for customer-specific extensions for multitenant SaaS applications. See Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application Extension 'Catering Management' on GitHub.

Estimate your cost-to-serve by calculating the total cost of ownership for multi-tenant SaaS applications built on SAP BTP. See Total cost of ownership calculation of multi-tenant SaaS applications built on SAP BTP on SAP Partner Portal.

Related Information

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

Tutorial and Reference Sample for a Cross-Stack Extension Application

URL: <https://help.sap.com/docs/btp/btp-developers-guide/building-blocks>

Title: *Explore | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Explore

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Explore

Developing a new application should always serve the purpose of your business, for example making a process more efficient and hence saving costs, making users more effective so they get better results, or supporting a business innovation which gives you a competitive advantage and allows you to make more money.

The explore phase is where the business opportunity is identified at a fairly high level. Typically key stakeholders are interviewed, or use case exploration workshops are conducted. The methods used differ, depending on whether you are looking to optimize existing processes or whether you are looking to innovate your business.

If you are looking to optimize existing processes, then you need to identify the business problem.

Identify the Business Problem

A business problem represents a gap between the desired and actual outcome of a task. It can be any hurdle, situation, or variation that prevents a business from achieving its objectives. A business problem may not have an immediately obvious resolution, and may be masked or hidden by perception and assumption. To define a true problem, you need to remove the factors that obscure the real reason behind the problem.

To solve a problem for an internal or an external customer when using a product, you need to ask questions that can help you understand the customer's needs, expectations, pain points, and feedback. Some examples of such questions are:

What are the tasks that customers are trying to achieve with a product?

How do the customers use the product in their daily workflow?

What are the main challenges or frustrations that the customers face when using the product?

How can you improve the product to better serve the customers' needs?

How satisfied are the customers with the product and the customer service?

What are the benefits or value that the customers get from the product?

How do the customers measure the success or impact of the product on their goals?

Create a Governance Model

One of the first and most important steps of your journey to the cloud is to establish an appropriate organizational setup and corresponding governance model. A clear and well-thought-out organizational setup makes it easier for your employees to adopt agile processes. See *Creating a Governance Model* in the SAP BTP Administrator's Guide.

Security Considerations in the Explore Phase

Addressing security in the explore phase when you're shaping the project vision with stakeholders lays a strong foundation for building secure, resilient applications. This early focus helps protect sensitive data, ensures compliance with regulations, and aligns security with business objectives from day one.

At this stage, you're not writing code yet, but you're making critical decisions that influence the security posture of your application. It's the right time to ask key questions, identify potential risks, and plan how to consider security aspects throughout the development lifecycle.

For example, a customer support team designing a ticketing system might identify risks such as data breaches if sensitive incident details or personal information are not properly handled. By proactively addressing these concerns now, you can avoid costly rework later and ensure compliance with data protection laws like the GDPR or HIPAA.

The following table summarizes the security guidelines you need to be familiar with:

Early-Stage Security Guidelines for Application Development

Security Guideline

Description

Define a Secure Software Development and Operations Lifecycle (Secure SDOL) Framework

Establish a secure software development lifecycle (SDLC) framework, using standards like OWASP to guide all phases.

See The Secure Software Development and Operations Lifecycle (secure SDOL) at SAP.

Engage and Train

Engage stakeholders to clarify their security expectations and ensure that development teams are trained on secure development principles. See Internal Wiki Page: SDOL-001: Security Training and Role Assignment.

Identify High-Level Security Risks

Conduct a preliminary risk assessment to identify high-level security risks based on the application use case. (For example, compliance, reputation).

Assess Potential Threats

Use threat modeling to identify potential threats relevant to your industry or app type, like SQL injection or insecure authentication.

See OWASP Threat Modeling Project.

Plan for Compliance

Determine regulations (for example, GDPR, HIPAA) that apply to your application data handling and storage needs.

Starting your application project with these steps helps ensure that security is considered from the beginning, making it easier to build a safe and compliant application.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/c2fec62b49fa43b8bd945c85ecc2e5bd.html>

Title: Development | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformDevelopment

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Development

Develop and run business applications on SAP Business Technology Platform (SAP BTP) using our cloud application programming model, APIs, services, tools, and capabilities.

SAP BTP supports multiple environments. It features a number of tools and programming languages that increase your flexibility and freedom when developing applications. You can choose the environment that best suits your use cases and skill set and get to work. You also have the possibility to integrate applications you've built with other solutions by SAP.

Environment options

Cloud Foundry

Kyma

ABAP

Benefits

Simplified developer experience for business application development

Large choice of programming languages

Intuitive “code-to-container” packaging and deployment, managed by the platform

Platform-managed application security patching and updates

Automatic application routing, load balancing, health checks, and multilevel self-healing

Support for CAP – an opinionated business app development framework

Take full advantage of the advanced features and rich ecosystem of Kubernetes

Free choice of programming languages and models (containerized deployments)

Combines microservices and serverless functions

Built-in, managed service mesh based on Istio, and other cloud-native open-source modules to reduce the development effort

Built-in, managed event mesh

Managed infrastructure: day-2 operations, security patches, and updates

Full administrator access

Refined horizontal and vertical automatic scalability

Dedicated application runtime

Zero downtime infrastructure setup by default

Support for CAP – an opinionated business app development framework

Support for on-premise connectivity

ABAP programming language

Fast prototyping with ABAP RESTful Programming Model (RAP)

Integrated development lifecycle

Reuse existing on-prem ABAP assets

Additional Information

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Development in the ABAP Environment

Shared Benefits

No infrastructure vendor lock-in

Build scalable multitenancy business applications (SaaS)

Out-of-the-box consumption of SAP and hyperscaler services

Built on industry standards and open technology

Good For

Managed build-on approach

Enterprise-grade business applications and services

Cloud-native web applications and services

Scalable, microservice-based applications

Small to medium extensions built with CAP/low-code tooling

Open build-on approach

Enterprise-grade applications

Cloud-native development of apps and services

Low latency infra-services communication

Reduced infrastructure management effort

Highly scalable, microservice-based applications

Applications built with CAP

User-centric process extensions

Robust, transactional cloud applications

Migrating and adapting add-ons to the cloud

Reusing existing on-premise ABAP code

Enabling ABAP developers to go to the cloud

Skills

Any major programming languages

SAP Fiori/UI5 and SAP HANA

Kubernetes knowledge

Docker

NodeJS or Python for serverless functions

Any major programming language

SAP Fiori/UI5 and SAP HANA

Ability to write modern ABAP code

Core data services

SAP Fiori/UI5 and SAP HANA

We provide detailed information about developing, configuring, and deploying your applications depending on your preferred environment and development approach.

Development in the Cloud Foundry Environment

Learn more about developing applications on the SAP BTP, Cloud Foundry environment.

Development in the ABAP Environment

Learn more about developing applications in the ABAP environment.

Development in the Kyma Environment

Learn more about developing applications in SAP BTP, Kyma runtime.

Related Information

SAP BTP Administrator's Guide

URL: <https://help.sap.com/docs/btp/btp-developers-guide/d04fc0e2ad894545aebfd7126384307c.html>

Title: Multitarget Applications in the Cloud Foundry Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Deploying to the Cloud Foundry EnvironmentMultitarget Applications in the Cloud Foundry Environment

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Multitarget Applications in the Cloud Foundry Environment

A multitarget application (MTA) is essentially a single application that consists of multiple parts. These parts are created using various technologies and share the same lifecycle.

The MTA developers outline the intended outcomes using the MTA model. This model consists of MTA modules, MTA resources, and the interdependencies between them. Afterward, the SAP Cloud Deployment service validates it, orchestrates the necessary actions, and automates the MTA deployment. The outcome is the formation of Cloud Foundry applications, services, and the generation of SAP specific content. For more information about the Multitarget Application model, see the official specification documents The Multitarget Application Model v.2 and The Multitarget Application Model v.3.

You can create and deploy a multitarget application in the Cloud Foundry environment as described below by following different approaches that can yield the same result:

Using SAP Web IDE Full-Stack as described in Developing Multitarget Applications - both the development descriptor `mta.yaml` and the deployment descriptor `mtad.yaml` are created automatically. The `mta.yaml` is generated when you create the application project, and the `mtad.yaml` file is created when you build the project.

Note

You may still need to edit the development descriptor.

Development descriptors are used to generate MTA deployment descriptors, which define the required deployment data. That is, the MTA development descriptor data specifies what you want to build, how to build it, while the deployment descriptor data specifies as what and how to deploy it.

Using Business Application Studio - the flow is similar to WebIDE but the development is done from BAS.

Using the Cloud MTA Build Tool. Afterward, you deploy the MTA using the Cloud Foundry Command Line Interface.

Note

An MTA development descriptor `mta.yaml` is required. You have to create it manually.

Manually - create the required files manually and deploy them using the Cloud Foundry Command Line Interface

Note

An MTA development descriptor `mta.yaml` is not explicitly required. In this case, a deployment descriptor is maintained instead.

Create the required files using an IDE of your choice.

To produce your custom artifacts, use a build technology of your choice.

Maintain your deployment descriptor document (`mtad.yaml`).

(Optional) If you want to have an MTA archive package, use the `mbt assemble`. For more information, see Cloud MTA Build Tool (MBT).

Use the `cf deploy` command to deploy the MTA package or directly from the build result directory.

To learn more about

See

Multitarget Application deployment descriptor

Defining Multitarget Application Deployment Descriptors for Cloud Foundry

Multitarget Application archive

Defining Multitarget Application Archives

Multitarget Application extension descriptor

Defining MTA Extension Descriptors

Multitarget Application structure

Multitarget Application Structure

How to deploy the Multitarget Application

Multitarget Application Plugin for Cloud Foundry Command Line Interface

Terms and Concepts

Terms and Concepts

Term

Description

Multitarget application (MTA)

An application comprised of multiple software modules, which are created with different technologies and deployed to different runtimes.

Development descriptor

A YAML file named `mta.yaml` that contains a list of all entities, such as modules, resources, and properties that belong to an application or are used by it at runtime, and the dependencies between them. It is automatically generated when an MTA project is created or modified, or when a module is added or removed. The developer needs to edit the descriptor manually to define resources, properties, and dependencies, as well as fill in missing information.

Deployment descriptor

A YAML file named `mtad.yaml` that contains a list of all entities which is created from either SAP Web IDE Full-Stack, SAP Business Application Studio, Cloud MTA Build Tool or manually. This file is similar to Development Descriptor but is used from the SAP Cloud Deployment service.

Module

A self-contained application of a certain type, which is developed, packaged and deployed.

Module type

A type that defines the structure and the development technology of a module. You can see a list of the module types at [Modules](#).

Resource

Any resource, such as an external service that is required by a module at runtime but not provided by the module itself.

Property

A property (key-value pair) of an application, module, or resource, that is used during deployment or at runtime.

Parameter

A reserved variable belonging to a module or resource, whose value is used during deployment or at runtime.

Dependency

A relationship between a module and another module, resource, or property, such as provides and requires.
provides: indicates the properties or parameters that are provided by a module or resource to other modules.
requires: indicates other modules or resources that are required by a module in order to run.

MTA archive (MTAR)

Archive containing a deployment descriptor, the module and resource binaries, and configuration files. The archive follows the JAR file specification.

Prerequisites and Restrictions

Prerequisites

You need the Space Developer role to deploy multitarget applications in the Cloud Foundry environment. For more information, see [About Roles in the Cloud Foundry Environment](#).

Restrictions

You have to consider the following limits for the MTA artifacts, which can be handled by the SAP Cloud Deployment service:

Maximum size of the MTA archive: 4 GB

Maximum size of MTA module content: 1 GB

Maximum size of MTA resource content: 1 MB

Maximum size of MTA descriptors (`mtad.yaml` and `MANIFEST.MF`): 1 MB

Limitations for transport scenarios via SAP Cloud Transport Management, where one transport import with multiple transport requests results into a single deployment process in SAP Cloud Deployment service with multiple MTAs:

The total size of all MTA archives in a single import request cannot exceed 20GB.

The maximum number of MTA archives used in a single import request cannot exceed 100.

Limitations for transport scenarios via the Enhanced Change and Transport System (CTS+), where one transport import with multiple transport requests results into a single deployment process in SAP Cloud Deployment service with multiple MTAs:

The total size of all MTA archives in multiple transport requests per one import cannot exceed 4GB.

The maximum number of MTA archives used in a single import request cannot exceed 500.

Unsupported Cloud Foundry Features

In addition to the features that are not supported by the Cloud Foundry environment on SAP BTP (See Supported and Unsupported Cloud Foundry Features), MTA deployment in Cloud Foundry currently does not support these features as well:

Sidecars. See <https://docs.cloudfoundry.org/devguide/sidecars.html>.

Route services. See <https://docs.cloudfoundry.org/services/route-services.html>.

Metadata. See <https://docs.cloudfoundry.org/adminguide/metadata.html>.

Per-route options. See <https://docs.cloudfoundry.org/devguide/custom-per-route-options.html>.

Note

These features might be implemented in the future. For the latest information, follow the release notes regarding Multitarget Applications for Cloud Foundry in What's New for SAP Business Technology Platform.

Related Information

JAR File Specification

Cloud MTA Build Tool

Release Notes for Multitarget Applications for Cloud Foundry

URL: <https://help.sap.com/docs/btp/btp-developers-guide/deliver>

Title: Deliver | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Deliver

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Deliver

Now that you have completed your design for the required user experience of your application along with the appropriate application architecture, it's time to deliver what you have designed. In other words, set up the development and test landscape, develop and test your application and finally release it to the productive environment. This applies to the first release of the application. From then on, you could apply a similar approach for subsequent releases, or preferably, move to a mode of continuous feature delivery, whereby additional features are researched with users, designed, developed and then deployed once ready, without having to be bundled in a new release of the entire application. See:

You start by setting up the landscape in SAP BTP. See Set Up.

Then, you continue with the development, choosing the programming language, runtimes, tools and others.

Have in mind that depending on the programming model, there are some specifics. See:

Develop

There are sample applications presented in tutorials and end-to-end scenarios, called missions:

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

Tutorials for SAP Cloud Application Programming Model

URL: <https://help.sap.com/docs/btp/btp-developers-guide/design-6bb733981248482e86559bc52a87aaff>

Title: Design | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Design

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Design

Good user experience (UX) goes beyond aesthetics and can have substantial business benefits, both financially and in overall efficiency. A great UX can improve productivity, enhance data quality, reduce training costs, and decrease the number of change requests.

A positive UX boosts user satisfaction, customer loyalty, and overall application adoption. Users are more likely to embrace an application that is intuitive and easy to use.

However, achieving a good UX involves overcoming several common challenges. Users often struggle with finding information, dealing with scattered data, excessive clicks, poor data entry interfaces, and unhelpful error messages. Beyond design, bad performance and unhelpful data visualization can also frustrate users.

The essential approach to designing a good application is design thinking, which emphasizes understanding users' needs through observation and feedback. Iteration is key: quickly sketch prototypes, gather user feedback, and refine your designs before moving to development. This ensures that the final product truly meets users' needs and optimizes resources effectively.

For organizations working with multiple applications, consistency is crucial. Users expect uniform visual design and interactions, easy access and navigation between applications, and common functions readily available. To achieve this, a design system is necessary. It ensures consistent user experience across various channels such as desktop, mobile, and so on.

In large-scale development, you also need robust frameworks, libraries, and user experience integration runtimes to manage hundreds of applications efficiently. Proper authorization tools are essential to control access for an extensive user base.

The SAP Fiori design system along with its enabling UI tools and technologies gives you the above: a comprehensive design system for web and mobile, along with a suite of development tools and UI frameworks, enabling SAP to provide a consistent UX with good usability across all SAP products, and enabling you to do the same for your applications. Using SAP's design system enables you to bring together business value, modern technology and excellent UX, thus benefitting business and users alike.

Find out more about user experience design and technology design in the next sections:

User Experience Design

Technology Design

Typical Development Use Case Patterns[^]

We recommend implementing the following use case patterns on SAP BTP:

Automate processes across backend systems

This pattern provides low code/no code capabilities to automate processes and is meant for business experts and developers.

Build web and mobile applications

This pattern provides mobile-native and web development capabilities and is meant for business experts and developers.

Develop full-stack applications

This pattern is meant for professional developers and has the following flavors:

Full-stack single-tenant applications

This pattern is meant for customers and implementation partners enabling them to develop full-stack applications on SAP BTP.

Full-stack multitenant Software-as-a-Service (SaaS) applications for partners that are independent software vendors

This pattern is meant for SAP partners enabling them to develop full-stack SaaS applications on the SAP BTP and to distribute these applications to their customers. Partners can easily onboard multiple customers (tenants) onto a single application with strictly separated data. This approach dramatically reduces the total cost of ownership at cloud scale.

The Poetry Slam Manager partner reference application on GitHub provides a comprehensive example to build, deploy, and run such a scalable SaaS solution. See [Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application 'Poetry Slam Manager'](#).

Tenant-specific extensions for partners and customers to add or modify features in multitenant SaaS solutions, such as the Poetry Slam Manager, without affecting the core product used by all tenants.

The Partner Reference Application Extension "Catering Management" demonstrates a robust framework for delivering highly-tailored functionalities for specific customers and industries, using SAP BTP and CAP extensibility as a foundation. See [Tutorials and Reference Sample for Custom Extension of a Full-Stack Multitenant SaaS Applications for Partners and Partner Reference Application 'Catering Management'](#).

CAP Design Principles[^]

SAP Cloud Application Programming Model (CAP) focuses on the domain of the application, by capturing

domain knowledge and intent instead of imperative coding. This means:

Close collaboration of developers and domain experts in domain modeling.

Out-of-the-box implementations for best practices and recurring tasks.

Platform-agnostic approach to avoid lock-ins, hence protecting investments.

Agnostic Design[^]

CAP avoids technology lock-ins /through /higher-level concepts and APIs, which abstract low-level platform features and protocols to a large extent. In particular, this applies to:

Platform-specific deployment approaches and techniques

Platform-specific identity providers and authentication strategies

Onboarding and offboarding of tenants in Software-as-a-Service (SaaS) solutions and tenant isolation

Synchronous protocols like /REST, /OData, or /GraphQL

Asynchronous channels and brokers like /SAP Event Mesh, /Message Queue, or /Kafka

Different database technologies including /SQL /and /NoSQL

These abstractions allow CAP to quickly adapt to new emerging technologies or platforms, without affecting the application code.

Open and Opinionated Design[^]

While CAP certainly gives opinionated guidance, it does this without sacrificing openness and flexibility. You as a developer stay in control of which tools or technologies to choose, or which architecture patterns to follow.

All abstractions follow a glass-box pattern that allows unrestricted access to lower-level things, if necessary.

Best Practices served out of the box with generic solutions for many recurring tasks.

Out-of-the-box support for SAP Fiori and SAP HANA.

Dedicated tools support provided in SAP Business Application Studio, and Visual Studio Code or Eclipse.

Domain-Driven Design[^]

CAP uses Core Data Services as its ubiquitous modelling language, with Core Data Service Aspects separating core domain aspects from generic aspects. Core Data Service's human-readable nature fosters collaboration of developers and domain experts.

As Core Data Service models are used to fuel generic providers — the database as well as application services — CAP ensures that the models are applied in the implementation. And as coding is minimized you can more easily refine and revise their models, without having to refactor large boilerplate code bases.

Security Considerations in the Design Phase[^]

In the design phase, you define the user experience and technical architecture of your application. This includes applying domain-driven design principles and following the best practices of the programming model you have chosen, CAP or ABAP Cloud, to create modular, user-centric solutions. Prioritizing security at this stage ensures your application is built on a resilient architecture that protects sensitive data and supports regulatory compliance.

Design decisions shape how users interact with your application and how its internal components communicate. For example, a retail company designing an e-commerce platform might implement a secure checkout process that enforces user authentication and encrypts payment data. By embedding security practices early—such as defining role-based access controls, securing APIs, and validating input—you reduce the risk of vulnerabilities being introduced into the system.

Proactively addressing security in the design phase not only streamlines compliance with regulations like the GDPR but also builds user trust and reduces the cost of future remediation.

The following table summarizes the security guidelines you need to be familiar with:

Designing Secure Applications from the Ground Up

Security Guideline

Description

Design Secure User Interfaces

Incorporate security features like secure authentication and input validation when designing the UI using SAP Fiori.

Define Access Control Models

Securely authenticate users and technical clients with methods such as OpenID Connect and OAuth. Design role-based access controls (RBAC) and attribute-based access control (ABAC) in domain models to restrict data access.

Secure API and Service Interactions

Design APIs with authentication (for example, OAuth) and encryption (for example, TLS) to protect data exchanges.

Plan for Secure Extensibility

Ensure extensions (for example, custom logic) are isolated and validated to prevent introducing vulnerabilities.

Validate Domain Models for Security

Review Core Data Services (CDS) models to ensure sensitive data is protected and access is restricted.

These guidelines help you to implement an application that integrates security into its UI, architecture, and domain models, supporting the development of a compliant and trustworthy solution.

See:

The Secure Software Development and Operations Lifecycle (secure SDOL) at SAP

SAP BTP Guidance Framework

URL: <https://help.sap.com/docs/btp/btp-developers-guide/dev-guide-operate-cap>

Title: *Establish End-to-End Observability* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Run and ScaleEstablish End-to-End Observability

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Establish End-to-End Observability

Introduction to Observability in SAP BTPà^

Observability in SAP Business Technology Platform (SAP BTP) empowers development and operations teams to proactively monitor, analyze, and optimize application performance using telemetry data - logs, metrics, and traces. It is structured into two complementary layers that together provide comprehensive visibility across your SAP landscape:

Central Observability, delivered via SAP Cloud ALM, offers a unified, strategic view across all SAP services and customer-built applications. It enables efficient issue detection through integrated monitoring, alerting, and business service management - helping teams maintain service reliability and performance.

Local Observability, enabled through tools such as SAP Cloud Logging, provides granular, technical insights into customer-built applications. It supports deep-dive analysis of runtime telemetry signals, facilitating rapid troubleshooting and performance optimization at the component level.

Both layers are interconnected through context-sensitive navigation and API integration, ensuring seamless transitions between high-level oversight and detailed diagnostics. Telemetry data—covering logs, metrics, and traces—flows from SAP BTP runtimes and services into these observability layers via OpenTelemetry-based instrumentation. This architecture reduces mean time to detect (MTTD) and mean time to resolve (MTTR), strengthening operational resilience and enabling faster, data-driven decision-making.

In addition, many SAP products provide embedded local monitoring capabilities for product-specific insights with minimal configuration. These capabilities are integrated into cross-navigation workflows from SAP Cloud ALM, allowing users to drill down from central dashboards into detailed product-level diagnostics for immediate action.

To complement these observability layers, the SAP Alert Notification service delivers real-time notifications for critical events across SAP BTP. It ensures that alerts generated in SAP solutions, central or local observability tools, can be routed to preferred communication channels - such as email, SMS, or third-party collaboration platforms - enabling proactive incident response and reducing operational risk.

SAP BTP spans Software-as-a-Service (SaaS) solutions like SAP S4/HANA, SAP HANA Cloud, SAP Analytics Cloud and more, and Platform-as-a-Service (PaaS) environments such as Cloud Foundry and Kyma. The BTP Observability capabilities integrate across these layers, ensuring consistent monitoring for both SAP-delivered and customer-built applications.

OpenTelemetry in SAP BTP: Strategy and Customer Valueà^

OpenTelemetry, the industry-standard for cloud-native observability, ensures consistent and scalable monitoring across distributed systems. It provides a unified set APIs, and SDKs to instrument applications and collect telemetry data - logs, metrics, and traces - across distributed systems. SAP has adopted OpenTelemetry to ensure consistent, scalable, and vendor-neutral observability across SAP-built and customer-developed applications on SAP BTP.

By integrating OpenTelemetry into SAP BTP runtimes, SAP enables customers to extend built-in observability

with their own instrumentation. This allows them to expose custom telemetry data for the workloads they manage, supporting deeper insights into application behavior and performance.

SAP's strategy centers on:

Standardized instrumentation: SAP and customer applications utilize the same OpenTelemetry-based Data Collection Libraries to provide telemetry signals consistently and openly.

Open API & integration: OpenTelemetry-compliant inbound APIs allow customers to connect non-SAP data sources, and outbound APIs enable integration with external observability tools.

Seamless data flow: SAP-built applications send telemetry data via a managed Data Collector Runtime, while customer-built applications transmit data directly to SAP Cloud ALM.

This approach empowers customers to build observability into their operational workflows. It supports a plethora of use cases around day-2 operations, incident management, and problem resolution. Customers benefit from SAP Cloud ALM's centralized monitoring and analytics capabilities. They can leverage deep-dive insights in local monitoring tools like SAP Cloud Logging, which support slicing and dicing of OpenTelemetry-based telemetry signals. These tools also offer flexibility to build their own operational workflows on top.

Use SAP Cloud ALM as a Centralized Operation Platform

SAP Cloud ALM, where ALM stands for application lifecycle management, is included in your SAP Cloud Service subscription containing Enterprise Support, cloud editions, in SAP Enterprise Support and in Product Support for Large Enterprises.

SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions. You benefit from an out-of-the-box, native cloud solution running on SAP BTP, designed as the central entry point to manage your complete SAP landscape with content-driven guided implementation and highly automated operations.

With the operations solutions in SAP Cloud ALM, you can handle both SAP BTP-based SaaS applications and your own applications.

SAP Cloud ALM for operations provides a personalized and unified operations user experience. You can use purpose-driven different monitoring, analysis, and management use cases optimized for certain operations aspects. SAP Cloud ALM for operations empowers you to understand the health of your SAP business solutions and acts as collaboration platform between lines of business and IT.

Under Supported Solutions, you see which monitoring capabilities are available with SAP Cloud ALM. Under SAP Solution, filter for SAP BTP and check the supported services from the operations area.

Benefit from the powerful Intelligent Event Processing, embedded analytics, and intelligence functionality with a large variety of inbound and outbound integration. Synthetic User Monitoring monitors the performance and availability of web applications from a client-side perspective based on Selenium-based scripts. Business Service Management consolidates all operational events at business service level and shows them in a well-arranged event calendar.

Specifically for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry runtime, there are the Data Collection Instrumentation Libraries based on OpenTelemetry designed to enable the data collection infrastructure in SAP Cloud ALM. With these libraries, you can collect data for the following use cases:

Real User Monitoring: Leverage both front-end and back-end measurements on single request level.

Health Monitoring: Gain insights on application health based on technical metrics.

Integration and Exception Monitoring: Collect and react on exceptions raised by the application.

Job and Automation Monitoring: Monitor SAP Job Scheduling Service jobs.

SAP Cloud ALM is the central place to integrate and access the monitoring of your existing IT service management processes, such as ticketing systems.

Inbound and outbound APIs enable you to integrate data from your own applications. For more information, see the API Guide for SAP Cloud ALM.

See:

Implement SAP Cloud ALM via SAP for Me

Application help for SAP Cloud ALM for Operations

SAP Cloud ALM for operations (SAP Support Portal)

Expert Portal for SAP Cloud ALM Operations (SAP Support Portal), with detailed configuration instructions

Observability in Action: SAP Cloud Logging for Problem Resolution and Analytics

SAP Cloud Logging is an instance-based observability service built on OpenSearch, designed to collect, store, and analyze logs, metrics, and traces from SAP BTP runtimes (such as Cloud Foundry, and Kyma) but also

arbitrary other contexts (support via API-based data ingestion). It supports OpenTelemetry standards and is integrated by SAP Cloud ALM, enabling developers and DevOps teams to gain deep operational insights for problem resolution tasks or build tailored observability workflows.

Telemetry signals can be leveraged for the following use cases for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry and Kyma runtime:

Log Analytics - Perform full-text search, filtering, and visualization of structured and unstructured log data to accelerate root cause analysis and detect anomalies across distributed systems.

Distributed System Monitoring - Understand request flows across microservices and correlate them with the Four Golden Signals (latency, traffic, errors, saturation) to provide end-to-end visibility for scaling and optimization decisions. Use correlation identifiers and access logs to analyze service interactions, dependencies, and system impact through dashboards and service maps. Monitor at different granularity levels, prioritize critical requests, and understand complete system call chains.

Technology Monitoring - Visualize runtime-specific metrics for Java and Node.js applications, including garbage collection, memory usage, and exceptions. Leverage OpenTelemetry signals to assess workload behavior and system health.

Custom Telemetry, Alerting & Dashboards - Instrument applications with service-specific logs, metrics, traces using OpenTelemetry and semantic annotations. Build custom dashboards and alerting rules to support proactive monitoring and incident response tailored to your business needs.

SAP Cloud Logging empowers teams with a robust observability foundation, enabling faster troubleshooting and resilient operations across SAP BTP workloads. Combining open standards, deep integration, and advanced analytics enables teams to move from raw telemetry to actionable insights - supporting resilient operations, faster troubleshooting, and more intelligent decision-making.

SAP Cloud Logging is built on top of OpenSearch. SAP proudly supports the OpenSearch Software Foundation as a premium member, actively helping to grow and maintain the open-source project. By engaging with this Linux Foundation initiative, SAP aims to nurture a thriving community and contribute to the future of innovative search solutions.

See:

Mission: Implement Observability in a Full-Stack CAP Application

SAP Cloud Logging at SAP Discovery Center

SAP Cloud Logging Product Page on SAP Help Portal

What Is SAP Cloud Logging?

URL: <https://help.sap.com/docs/btp/btp-developers-guide/develop-7e306865dadb4473a4d66d81b7d83004>

Title: Develop | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...DeliverDevelop

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Develop

When using SAP Cloud Application Programming Model (CAP), you can choose the right runtime of your use case: SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime. To make an informed decision, see:

SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Metering:

Cloud Foundry runtime: Metering starts when you start using the runtime memory, for example, when you deploy an application.

If your global account uses the consumption-based commercial model and you want to understand how the consumption of SAP BTP, Cloud Foundry runtime is calculated, see Consumption Monitoring in the SAP BTP, Cloud Foundry runtime documentation.

Kyma runtime: Metering starts when you enable the Kyma runtime. This triggers the creation of a dedicated Kubernetes cluster, and the cluster is metered from the start, before you have deployed any applications.

For Cloud Foundry , we recommend that you create multitarget applications for managing dependencies more easily. For more information, see Using Multitarget Applications to Manage Dependencies.

For best practices, guidelines and step-by-step instructions, see Development.

Development Options Overview

The following graphic is designed to help you find the information you need for your programming purposes. The bottom row represents the tools, frameworks, services, and deployment options recommended by SAP. If you want full flexibility you can also bring your own development tools and languages, as shown in the top row. This image is interactive. Hover over each area for a description. Click highlighted areas for more information.

Development Guidelines

To ensure consistency and foster collaboration between developers, we recommend that you define guidelines that include standards for programming principles, code styles, and naming conventions.

SAP provides the following official guidelines:

SAPUI5 Guidelines

SAP Fiori Design Guidelines

SAP BTP and Third-Party Cookies

The deprecation of third-party cookies by browser manufacturers might impact applications running on SAP BTP. For example, logon scenarios in your application might fail due to session management and authentication issues.

Third-party cookies are set by domains other than the one a user is directly visiting. They are commonly used for online advertising and tracking user behavior across multiple websites. In SAP BTP, SAP uses third-party cookies for service management and single sign-on.

Prepare your applications for the third-party cookie deprecation and test them. See [Preparing and Testing Your Solution for Third-Party Cookie Deprecation](#).

All the available solutions are listed in the [SAP Note 3409306 - Removal of Third-Party Cookies in Google Chrome and Microsoft Edge Browser](#).

Tutorials and End-to-End Scenarios

There are tutorials and end-to-end scenarios developed to show you how to apply the recommended path for the Cloud Application Programming Model (CAP). All these tutorials and scenarios are based on the Incident Management sample application. For a full list of the tutorials and their use cases, see [Tutorials for SAP Cloud Application Programming Model](#).

Related Information

[Programming Languages, Tools, and APIs](#)

[Security Considerations for Applications](#)

[Developing Resilient Applications](#)

[Using Multitarget Applications to Manage Dependencies](#)

[Extending Existing SAP Solutions Using SAP BTP](#)

[Performing UI, Usability, and Unit Tests](#)

[Connecting to Remote Systems](#)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/developing-resilient-applications>

Title: [Developing Resilient Applications | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...DevelopDeveloping

Resilient Applications

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Developing Resilient Applications

Our best practices about resilient application design help you to make your applications running on SAP BTP stable and highly available.

There are many different principles and patterns you can use to make your software resilient. However, it isn't always easy to find the combination that best fits your applications. For an overview of the various options you have when developing applications and detailed information about the individual patterns you can use, as well as examples for applied resilience principles and how to develop your own resilient apps on SAP BTP, read [Developing Resilient Apps on SAP BTP](#).

For examples on how to use selected SAP BTP services in a multi-region setup, see [Route Multi-Region Traffic to SAP BTP Services Intelligently](#).

You can also use the [Architecting Multi-Region Resiliency for SAP BTP Use Cases](#) reference architecture in the SAP Discovery Center. It shows the multi-region high availability architecture for SAP BTP services and the

applications built on SAP BTP.

Developing Resilient Applications in Cloud Foundry[^]

In SAP BTP, Cloud Foundry environment, you can use the availability zones: Benefit from the high availability mechanisms in Cloud Foundry by setting up your applications with multiple instances. You do not have to do anything explicitly to distribute the instances across the different availability zones - this is handled by the platform. For more information, see [Availability Zones in the Cloud Foundry Environment](#).

Setting up your application with multiple instances might have an impact on your applications' capability for handling requests: In case of an availability zone failure in a 3-availability-zone-deployment, at least one third of the application instances are affected and unavailable until Cloud Foundry can reschedule these instances on a healthy virtual machine.

During that period, the remaining instances in the healthy availability zones have to carry the request load.

Distribution During an Availability Zone Failure

Developing Resilient Applications in Kyma[^]

Learn how to develop stable Kubernetes and microservice-based applications, as well as Istio-related resiliency tips with [Develop Resilient Applications in the Kyma Runtime](#).

Related Information[^]

[Resilience, High Availability, and Disaster Recovery \(Cloud Foundry, ABAP and Kyma Environments\)](#)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/discover>

Title: [Discover | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Discover

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Discover

Now that the business opportunity has been identified by the explore phase, you can start to discover the concrete needs of your users as well as getting an understanding of the available technology for your landscape.

Identifying Required Use Cases[^]

Once you have an understanding of the business opportunity you want to address, you next will need to understand in detail what your end user's tasks and challenges are. You need a holistic view of the business roles of your application's users, their jobs to be done, the information they need to get them done, along with their needs and wishes.

To get the relevant insights, you will need to visit your customers onsite or remotely, speak to everyone involved, and if onsite, observe users doing their daily work. You should gain a clear understanding of all the business roles involved, their typical tasks and pain points, and the sequence of activities.

The goal of the discover phase is to gain a thorough understanding of the user's needs, independent of any existing application or technology. In other words, which information do they need? When do they need it? What business outcomes do they need to achieve?

By the end of the discover phase, you should know what needs to be achieved – without knowing how you want to achieve it, that is addressed in the design phase.

For more information about user research during the Discover phase, have a look at the [User Research Resources](#). The methods outlined there contain the most commonly practiced user research methods at SAP.

Security Considerations in the Discover Phase[^]

Addressing security during the discover phase—when you're gathering detailed requirements and exploring technical options—ensures that protection is a core part of your application's foundation. This phase is about understanding user needs, assessing feasibility, and building early prototypes, making it the ideal time to identify potential risks and plan for secure implementation.

At this stage, you define your application's features and technical requirements. For example, a retail company planning an e-commerce application must ensure secure payment APIs are in place to protect transaction data—an early decision that directly impacts the system's architecture. By evaluating third-party components, identifying data flows, and conducting early security testing on prototypes, you can uncover vulnerabilities before development begins. This proactive approach saves time, reduces cost, and supports compliance with regulations like the GDPR.

The following table summarizes the security guidelines you need to be familiar with:

Security in the Discover Phase: From Requirements to Risk Reduction

Security Guideline

Description

Map Data Flows for Security

Identify and document how sensitive data moves through the system, including user inputs, API calls, and database interactions, to identify potential security risks. You can use OWASP Threat Dragon to create data flow diagrams for this purpose.

Establish Secure Architecture Principles

Define architectural guidelines to help ensure that the system is built with security in mind, such as microservice isolation, the principle of least privilege, and multi-layered security approach (defense-in-depth).

Validate Third-Party Service Security

Review the security certifications and practices of third-party services (for example, APIs, libraries) you integrate, and align with SAP BTP Security recommendations.

Incorporate Security in Prototypes

Include basic security features, like input sanitization or mock authentication, in your prototypes to test their effectiveness early.

Plan for Data Security During User Research

Protect user data collected during research (for example, interviews, observations) by anonymizing inputs and securing storage.

Understand Boundary Conditions

Understand boundary conditions for the new application, like integration into the existing system landscape (regarding IAM, Audit log).

By implementing these guidelines in the discover phase, you improve the security of your application, laying the groundwork for a resilient and compliant application. See [Setting Up Your Security and Compliance Model](#).

URL: <https://help.sap.com/docs/btp/btp-developers-guide/extending-existing-sap-solutions-using-sap-btp>

Title: *Extending Existing SAP Solutions Using SAP BTP | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...DevelopExtending Existing SAP Solutions Using SAP BTP

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Extending Existing SAP Solutions Using SAP BTP

SAP BTP offers services, tools, and capabilities to develop, extend, or integrate business applications in the cloud. You can implement additional workflows or modules on top of existing solutions, which lets you benefit from out-of-the-box security, inherited data access governance, user interface embedding, and so on.

Overviewà

All standard SAP solutions are offered with customizing capabilities. Additionally, customers often have their own requirements for innovative or industry-specific extensions and the extension capability of SAP BTP can help them build, deploy, and operate their new functionalities easily and securely.

You can use SAP BTP to extend standard SAP solutions without disrupting their performance and core processes. When building extension applications, you can also benefit from the automation of the integration between SAP BTP and the extended SAP solutions.

There are three main extension types you can use to extend your SAP solution:

In-app extensions: allows you to extend the back end and the UI directly in your SAP solution. You can use in-app extensions only if the SAP solutions that you want to extend provide such a possibility. For example, you can use in-app extensions for extending SAP SuccessFactors, and SAP S/4HANA Cloud. See:

[SAP S/4HANA Cloud In-App Extensibility](#)

[SAP SuccessFactors Extension Center: In-App Extensions](#)

Side-by-side extensions: allows you to integrate with your SAP solution additional business functionality deployed in SAP BTP.

The advantages of the side-by-side extensions can be grouped in these categories:

Business: you can include the access of new user groups, you can extend the scope with custom solutions, you can complement data and business services from different back ends.

Technology: more developers are enabled, your digital core remains focused, you can add innovative solutions by leveraging new technology services in the cloud.

Operations: you can share the extensions

When planning your extension, consider these recommendations:

Always start simple to check the result quickly. Later on, you can grow the complexity of your extension slowly.

Keep your digital core focused and bring to SAP BTP only the data that makes sense.

Define what you want to achieve in the beginning and look for the easiest way to do that. For example, if you want to add a field in SAP S/4HANA, it's better to create an in-app extension, than to implement an extension application on SAP BTP.

Ensure the availability of suitable APIs and events, because you will need them when developing and extension application on SAP BTP.

In SAP BTP, you have these options to extend your SAP solution:

Extensions with automated configurations in the Cloud Foundry and Kyma environments: applicable for SAP S/4HANA Cloud, and SAP SuccessFactors.

Extensions with automated configurations in the Kyma environment: applicable for SAP S/4HANA Cloud, SAP SuccessFactors, SAP Cloud for Customer, SAP Commerce Cloud, and SAP Field Service Management.

Extensions with Automated Configurations

SAP BTP provides a standard way for extending SAP solutions and developing event-driven extensions and applications. This framework includes:

Simplified, standardized and unified extensibility and configuration for the SAP solutions.

Central catalog per customer for all connected SAP systems where data such as APIs, events, credentials and other is stored. You can benefit from business services and actionable events across end-to-end business processes.

If you have to group the systems of different SAP solutions in the same business case, you can set up the connectivity between all these systems and SAP BTP in a single formation.

The following SAP solutions currently support the automated configurations:

SAP S/4HANA Cloud

SAP SuccessFactors

SAP Cloud for Customer

SAP Commerce Cloud

SAP Field Service Management

Use Cases¹

The extension use cases include but are not limited to:

Extending the user interface of your SAP cloud solution.

You can complement an existing SAP solution with a new or improved SAP Fiori user interface without adding any major logic or data.

You can also provide necessary information and tools on mobile devices.

For example:

Building completely new user interfaces that can be seamlessly integrated in the SAP SuccessFactors Employee Central home page.

Adding new functionality to SAP SuccessFactors by connecting it to other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP S/4HANA Cloud.

Building completely new SAP Fiori-based user interfaces that can be seamlessly integrated in the SAP Fiori launchpad.

Enhancing the functionality of your SAP cloud solution by connecting it with other SAP solutions.

For example:

Connecting SAP S/4HANA Cloud with other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP SuccessFactors.

Connecting SAP SuccessFactors by connecting it to other SAP solutions, such as SAP Commerce Cloud, SAP Cloud for Customer, SAP Field Service Management, SAP S/4HANA Cloud.

Enhancing core business processes with analytics and machine learning capabilities.

Extend data insights by consolidating and combining data in one central place.

Building an extensions ecosystem, by creating and operating SaaS applications and selling them to multiple customers.

For example, as an SAP Partner, you can build a multi-tenant SaaS extension application and provide it to your customers via:

SAP App Center
SAP BTP cockpit
Resources^

The following resources are available for your extension scenarios. The links to various documents guide you through the configuration tasks that you need to perform to enable the SAP BTP for developing extension applications for your existing SAP solutions, and the learning journeys are collections of links to additional resources.

SAP S/4HANA Cloud

Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment (SAP Help Portal)
Extend SAP S/4HANA Cloud on SAP Business Technology Platform (Mission in Discovery Center)
SAP Extensibility Explorer for SAP S/4HANA Cloud

Tutorials about the integration of SAP BTP ABAP Environment with SAP S/4HANA Cloud

SAP Customer Experience (CX) products

Extending SAP Customer Experience Products in the Kyma Environment (SAP Help Portal)

Create Extensions in the Kyma Environment Using Functions (SAP Help Portal)

SAP SuccessFactors

Extending SAP SuccessFactors in the Cloud Foundry and Kyma Environment (SAP Help Portal)

Extend SAP SuccessFactors on SAP BTP (Mission in Discovery Center)

Build Resilient Cloud Native Applications (Mission in Discovery Center)

Related Information^

Extensions

Extension Architecture Guide

SAP Application Extension Methodology Overview

URL: <https://help.sap.com/docs/btp/btp-developers-guide/from-concept-to-code>

Title: Technology Design | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...DesignTechnology Design

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Technology Design

Before starting to implement an application on SAP BTP, it's better if you do some planning and research beforehand. See:

Explore

Discover

Deliver a Cloud Application from Concept to a Final Product^

The columns in the following graphic show for which use cases using a well-modularized architecture suits best, what is the motivation, the steps and the benefits.

The image is interactive: you can click the column headings to take you to the relevant part of the relevant section below.

Context

Let's assume the following scenario: you and your team are tasked to deliver a multitenant application on SAP BTP or SAP S/4HANA Cloud Public Edition and you want to base the application either on SAP Cloud Application Programming Model (CAP) or on ABAP Cloud.

Trigger

Use the following triggers as a motivation to engage in sophisticated design process:

Renovating, evolving or reimplementing an existing solution

This involves transforming a legacy system into a modern, more efficient and scalable solution. Legacy systems often pose difficulties in terms of maintainability, integration with newer technologies, and can be a bottleneck for a business's growth. This transformation process could involve refactoring code, adopting new architecture patterns like microservices, or completely reimplementing the software with a new technology stack.

Designing a new Software-as-a-Service solution based on microservices

Software-as-a-Service is a model where software is provided over the internet on a subscription basis. When designing a new Software-as-a-Service solution based on microservices, the system is decomposed into a collection of services that are independently deployable, scalable, and can be developed by different teams. Each microservice corresponds to a specific business capability and can be developed, deployed and scaled independently. This architectural style promotes flexibility and scalability, and it can speed up the development process by enabling parallel development across teams. It's important to design these services around the business capabilities.

Considering more asynchronous behavior and event-sourced applications

Asynchronous behavior in an application refers to operations that can be executed independently and don't block the flow of the program. This makes the application more responsive and can improve system-wide performance. Event sourcing is an architectural pattern that stores changes to the application state as a sequence of events, rather than just storing the current state. This can provide a full audit trail, simplify debugging by allowing the state of an application at any point in time to be recreated, and can facilitate asynchronous, message-driven architectures. By integrating these into the system, it can enhance the system's scalability and reliability.

Designing a Software-as-a-Service application on SAP BTP

When designing a Software-as-a-Service application on SAP BTP, the application is built to be delivered over the internet and leverages the services provided by SAP BTP. This can include data storage services, AI and machine learning capabilities, analytics and more. Using such a platform can speed up development time, as many of the underlying infrastructure considerations and services are taken care of by the platform.

Steps

Follow these steps to ensure smooth implementation of your application:

Understand the business domain.

Understanding the business domain is of the utmost importance. Hence, before starting your development, make sure that you understand the business opportunity outlined in the Explore phase, and that you have completed the Discover phase so that you have a thorough understanding of the user needs and required use cases. This will ensure that you create software that is useful, flexible, and capable of evolving alongside the business.

Identify the required modules.

Identifying the required modules means breaking down the overall system into smaller, manageable, and functionally cohesive units. Each module is designed to encapsulate a specific portion of the domain model and is typically aligned with a specific sub-domain or business capability. By identifying the modules, developers can ensure separation of concerns and make the system more maintainable. Each module should be able to operate independently, reducing dependencies and allowing for parallel development and easier testing.

Define the communication between the modules.

Once the modules are identified, the next step is to define how they will interact with each other. This involves establishing interfaces, contracts, or APIs for communication. This is often guided by the concept of context mapping, which helps to define the relationships and interactions between different related contexts.

Communication between modules should be carefully managed to preserve the autonomy of the modules. The communication could be synchronous or asynchronous, and could be based on various patterns like request/response, events, or commands.

Plan the implementation.

After you have an understanding of the business domain, have identified the required modules, and defined how they will communicate, the next step is to plan the implementation. This is the moment where you decide which technologies, patterns, and practices to use to build each module. This can include selecting the programming languages, databases, message brokers, and other tools that will be used. It also involves choosing the architectural patterns (like layered architecture, hexagonal architecture, or microservices) and design patterns that will be applied. The implementation should be guided by the principles of the domain model, encapsulating the business logic in entities, value objects, aggregates, and domain events. Additionally, the implementation should consider non-functional requirements like scalability, performance, security, and maintainability.

Value

If you follow the steps mentioned in the previous section, you will achieve the following benefits:

Well-modularized architecture

In the context of software development, a well-modularized architecture refers to the design of a system where the different functionalities and responsibilities are partitioned into distinct modules. Each module encapsulates a specific subset of the system's behavior and can be developed, tested, and maintained independently of the others. Modules are often designed to be loosely coupled, meaning that they interact with each other through well-defined interfaces, but their internals are hidden from each other. This encapsulation increases the maintainability of the system because changes inside one module don't impact the others. It also improves the scalability of the system, as individual modules can be scaled up or down based on their individual requirements. Finally, modularization can increase the speed of development because different teams can work on different modules at the same time without interfering with each other.

Higher likelihood to deliver value to your customers

The focus is on understanding the business domain and creating a software model that reflects and serves that domain. By aligning the software closely with the business needs, there's a higher likelihood of delivering value to the customer. This comes from a deep understanding of the business, its rules, and its needs. This understanding guides the software development process, ensuring that the software solves the right problems and provides the features that the business actually needs. As a result, the software is more likely to be valuable to the end users, who are the ones interacting with the business domain on a daily basis.

Better decomposition that would lead to shorter innovation cycles and clearer focus

Decomposition in software design is the process of breaking down a complex system into smaller, manageable parts or components. Better decomposition, often achieved through practices like domain-driven design and microservices architecture, allows for clearer focus on individual parts of the system and can lead to shorter innovation cycles. With a well-decomposed system, teams can focus on specific components or services, understand them thoroughly, and make improvements or innovations without having to understand the entire system. This can significantly shorten the time it takes to go from an idea to a working feature, because changes are localized to specific parts of the system and don't require broad, system-wide understanding or modifications. Moreover, better decomposition can lead to clearer focus because each component or service can be designed around a specific business capability or domain concept. This allows the team working on that component to focus on the specific business problems and user needs related to that capability or concept, rather than being distracted by other parts of the system. This kind of focus can lead to better solutions and more valuable software.

Understand Modularization in General

Modularization is a general programming concept where you separate program functions into independent pieces, called modules. These modules then act like building blocks, with each block containing all the necessary parts to execute a piece of the functionality. When the modules are put together, they make up the executable program.

A modular code is:

Easier to maintain

It's easier to read, understand, debug, and modify code that is modularized. It also reduces the risk of introducing errors or conflicts when changing or adding new features. A good modularization also enables reuse of existing modules across different products, which saves time and effort.

Easier to test

Each module can be tested individually and independently from the rest of the system. This also facilitates automated testing and continuous integration, which improve the quality and reliability of the software product.

Easier to scale

Each module can be deployed, updated, or replaced separately, without affecting the whole system. This also allows for parallel development and faster delivery of new features or bug fixes. A good modularization also supports distributed and cloud-based architectures, which enhance the performance and availability of the software product.

More reliable

Each module has a clear and well-defined interface and responsibility. This also reduces the coupling and dependency between modules, which minimizes the impact of failures or errors in one module on the rest of the system. A good modularization also enables fault tolerance and recovery mechanisms, which improve the resilience and robustness of the software product.

More adaptable

Each module can be easily modified or extended to meet changing requirements or customer needs. This also

increases the flexibility and diversity of the software product, as different modules can be combined or configured in various ways.

Easier to evolve

Each module can be evolved independently or collaboratively, without affecting the overall functionality. This also promotes innovation and creativity, as new modules can be created or reused to generate novel solutions or features.

Understand Modularization in the Context of Microservice-Based Architecture

The right modularization is very important in the context of microservices, because microservices are essentially a form of modular software development. Microservices aim to create small, focused, and independent services that communicate through well-defined interfaces and protocols. Each service is responsible for a specific functionality or domain, and can be developed, deployed, and scaled independently from other services. This way, microservices enable a high degree of modularity, flexibility, and agility in software development.

However, not all modularizations are equally effective or beneficial for microservices. A poor modularization can lead to problems such as:

High coupling and low cohesion

If the services are not well-defined and separated, they may depend on each other too much, or have too many responsibilities. This can result in tight coupling between services, which makes them hard to change, test, or reuse. It can also result in low cohesion within services, which makes them complex, inconsistent, or redundant. High coupling and low cohesion reduce the maintainability, testability, and reliability of microservices.

Service sprawl and fragmentation

If the services are too fine-grained and numerous, they may become difficult to manage, monitor, and coordinate. This can result in service sprawl, which increases the operational complexity and overhead of microservices. It can also result in service fragmentation, which reduces the performance and availability of microservices. Service sprawl and fragmentation reduce the scalability and adaptability of microservices.

Lack of alignment and consistency

If the services are not aligned with the business domain and goals, they may not deliver the expected value or quality, which reduces the relevance and effectiveness of microservices. It can also result in lack of consistency, which reduces the usability and interoperability of microservices. Lack of alignment and consistency reduce the evolution and innovation of microservices.

The right modularization is essential if you want to benefit from using microservices. For example:

Maintain modules easily

The right modularization helps you create simple, consistent, and reusable services that are easy to read, understand, debug, and modify. It also enables the isolation and decoupling of services, which reduces the risk of errors or conflicts when changing or adding new features.

Test each service individually and independently from the rest of the system

The right modularization enables you to use automated testing and continuous integration, which improve the quality and reliability of the software product.

Deploy, update, or replace each service separately, without affecting the whole system

The right modularization supports distributed and cloud-based architectures, which enhance the performance and availability of the software product.

Have clear and well-defined interface and responsibility of each service

The right modularization reduces the coupling and dependency between services, which minimize the impact of failures or errors in one service on the rest of the system. It also enables fault tolerance and recovery mechanisms, which improve the resilience and robustness of the software product.

Modify or extend each service to meet changing requirements or customer needs

The right modularization increases the flexibility and diversity of the software product, as different services can be combined or configured in various ways.

Evolve each service independently or collaboratively, without affecting the overall system functionality

The right modularization promotes innovation and creativity, as new services can be created or reused to generate novel solutions or features.

Learn Why Domain-Driven Design Is an Adequate Approach

Domain-driven design is a major software design approach, focusing on modeling software to match a domain according to input from that domain's experts. Under domain-driven design, the structure and language of

software code (class names, class methods, class variables) should match the business domain.

Domain-driven design helps achieving the traits of a good modularization by:

Placing the project's primary focus on the core domain and domain logic

Domain-driven design emphasizes close collaboration between domain experts and developers to create software that accurately models the problem domain. By focusing on the domain's core concepts, relationships, and behavior, domain-driven design enables you to create more robust, flexible, and maintainable systems.

Building complex designs on the model of the domain

Domain-driven design helps you to define the boundaries of each module by identifying the core concepts, aggregates, and their associated business logic. This enables teams to develop and maintain these modules independently, promoting scalability and modularization within the overall system.

Initiating a creative collaboration between technical and domain experts to iteratively refine a conceptual model that addresses particular domain problems

Domain-driven design fosters a common language shared by domain experts, users, and developers: the ubiquitous language. The ubiquitous language is used in the domain model and for describing system requirements. This enhances the communication and understanding among the stakeholders, and facilitates the adaptation and evolution of the software product.

Learn About the Domain-Driven Design Starter Modeling Process[^]

The domain-driven design starter modeling process is a process for learning and applying domain-driven design in practice. It covers eight steps from aligning with the business model to coding the domain model. It is flexible and iterative, and suitable for beginners who want to master domain-driven design. See the Domain-Driven Design Starter Modeling Process GitHub repository that is part of the Domain-Driven Design Crew GitHub organization.

Check How Complex Your Scenario Is[^]

Check which requirements your scenario has and sum up the points for the respective requirement. If your score is 7 or higher, you should evaluate how to leverage the concept and tools of the domain-driven design in your project.

We also recommend that you follow the domain-driven design modeling process.

See Domain-Driven Design Starter Modeling Process.

Scenarios Ordered by Complexity

Scenario Requirements

Points

Comments

For applications that work with the four functions: create, read, update, delete (CRUD), and only perform simple database queries, no special design is required. A simple database tool with a user interface suffices.

0

A simple application does have the complexity to benefit elaborated design methods.

You have a simple application if it has up to 30 use cases. You can build it with a CAP application that works with the four functions: create, read, update, delete (CRUD).

1

Just for reference, this means your application does not have more than 30 functions.

Your application has between 30 and 40 use cases.

2

Make sure you don't underestimate any hidden complexities.

However, having many methods or functions doesn't always mean high complexity. They might be simple and well-structured. It's when a class or module has many responsibilities, indicated by methods or functions that perform widely varying tasks, that complexity becomes a challenge.

Your application may not be complex now, but it could grow later.

3

Talk to domain experts to discuss the complexity of your application. The complexity increases with the number of features that the stakeholders need.

Your application will change a lot over time, and you don't know what kind of changes will happen.

4

You require a sophisticated design method to manage the complexity of refactoring your model over time.

You are building an application in a new and exciting domain that nobody has done before. This means it could

be complex and challenging.

5

You need to collaborate with domain experts and try different models to make your application work.

Related Informationà

Domain-Driven Design Crew GitHub Organization

Domain-Driven Design Community at SAP

URL: <https://help.sap.com/docs/btp/btp-developers-guide/mission-develop-enterprise-grade-cap-application-following-sap-btp-developer-s-guide>

Title: Mission: Develop an Enterprise-Grade CAP Application | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming ModelMission: Develop an Enterprise-Grade CAP Application

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Mission: Develop an Enterprise-Grade CAP Application

You can develop enterprise-grade CAP applications and deploy them in SAP BTP. The Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide mission includes options for tracking changes, adding audit logs, and uploading attachments.

See Develop an Enterprise-Grade CAP Application Following the SAP BTP Developer's Guide.

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/mission-develop-multitenant-cap-application>

Title: Mission: Develop a Multitenant CAP Application | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming ModelMission: Develop a Multitenant CAP Application

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Mission: Develop a Multitenant CAP Application

SAP BTP provides a multitenant functionality that allows application providers to own, deploy, and operate tenant-aware applications for multiple consumers, with reduced costs. For example, the application provider upgrades the application for all your consumers instead of performing each update individually, or share resources across multiple consumers. The application consumers launch the applications using consumer-specific URLs, and configure certain application features.

With tenant-aware applications, you can:

Separate data securely for each tenant

Save resources by sharing them among tenants

Update applications efficiently, in a single step

A business solution is typically a multitenant application that consists of UI modules, business logic and a connection to a database. Unlike other applications that are integrated into SAP Build Work Zone, it also contains the site design-time definitions such as tiles, roles, spaces, and pages, and the site entity that wraps in all together. You can develop business solutions to easily share them with consumers. There are two types of business solutions you can choose from: a central or local entry point. These are the main differences between the central and local entry point:

Key Differences Between the Approaches Supporting Multitenancy in SAP Build Work Zone

Feature

Central Entry Point

Local Entry Point

Consumer-side SAP Build Work Zone service instance

Required: standard or advanced service plan

Not required

App router management

Managed app router: you get it out of the box

Standalone app router: extensible

Site administration

Administered by the consumer: full SAP Build Work Zone service instance in the consumer subaccount

Provider-managed: not administered by the consumer

Layout customization

Customizable by the consumers: the administrator defines the spaces/pages with federated business application

Layout is fixed by the provider, groups are not supported

You can find a detailed comparison between central and local entry point at Creating Business Solutions.

To try out the multitenant functionality on the Incident Management sample application, see Develop a Multitenant CAP Application Following the SAP BTP Developer's Guide.

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/mission-develop-side-by-side-cap-based-extension-application>

Title: [Mission: Develop a Side-by-Side CAP-Based Extension Application | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming ModelMission: Develop a Side-by-Side CAP-Based Extension Application

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Mission: Develop a Side-by-Side CAP-Based Extension Application

Using the SAP BTP Developer's Guide you can implement side-by-side SAP Cloud Application Programming Model (CAP)-based extension applications and deploy them in SAP BTP. The Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide mission includes tracks for a remote service connectivity and eventing. In the scenario, an SAP S/4HANA Cloud system is used. If you don't have such a system and you still want to try out the mission, you can use a mock server instead. Depending on what you choose, there are 8 different options:

Connect to a remote service

Option 1a: SAP BTP, Cloud Foundry runtime and mock server

Option 2a: SAP BTP, Cloud Foundry runtime and SAP S/4HANA Cloud

Option 3a: SAP BTP, Kyma runtime and mock server

Option 4a: SAP BTP, Kyma runtime and SAP S/4HANA Cloud

Implement eventing

Option 1b: SAP BTP, Cloud Foundry runtime and mock server

Option 2b: SAP BTP, Cloud Foundry runtime and SAP S/4HANA Cloud

Option 3b: SAP BTP, Kyma runtime and mock server

Option 4b: SAP BTP, Kyma runtime and SAP S/4HANA Cloud

Out of these 8 options, you can choose one option for the remote service and the corresponding option for the eventing. If you are interested in the remote service connectivity, you don't have to finish the eventing option as well. However, if you are interested in the eventing part, you have to finish the respective remote service connectivity option.

See Develop a Side-by-Side CAP-Based Extension Application Following the SAP BTP Developer's Guide.

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/mission-implement-observability-in-full-stack-cap-application>

Title: [Mission: Implement Observability in a Full-Stack CAP Application | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming ModelMission: Implement Observability in a Full-Stack CAP Application

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Mission: Implement Observability in a Full-Stack CAP Application

You can add observability to the Incident Management application. Observability helps make informed decisions using real-time data like logs, metrics, traces, and events. It provides insights into SAP and customer systems to improve efficiency, scalability, resilience, and availability.

For detailed observability data, use SAP Cloud Logging. SAP Cloud Logging allows you to analyze logs, metrics, and traces to understand the performance of your SAP BTP workload in the SAP BTP, Cloud Foundry, and SAP BTP, Kyma environments.

See [Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide](#).

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/performing-ui-usability-and-unit-tests>

Title: [Performing UI, Usability, and Unit Tests | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...DevelopPerforming UI, Usability, and Unit Tests

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Performing UI, Usability, and Unit Tests

You should always conduct careful testing to ensure that your application is of high quality. Create a release candidate to propagate throughout your landscape.

For example, consider testing the user interface, usability, and performance, as well as running general unit tests. Developers should use unit tests to ensure that software modules behave as expected. Unit tests are the smallest tests, and they detect issues fast. Taking good care of unit tests usually leads to better maintainable and understandable code. If you use the Continuous Integration process, you should consider automated tests as part of the pipeline.

SAPUI5à^

SAPUI5 supports QUnit testing. We recommend that you write unit tests as small as possible, ensuring that you only test one single thing at a time. This will not only make debugging easier, but also maintaining the tests in the long run.

For an overview of available testing options for SAPUI5 developments, see [Testing and Performance Measurement](#).

For SAPUI5 and SAP Fiori testing, please see the demokit for SAP UI5: <https://sapui5.hana.ondemand.com/#/> topic.

Testing in SAP Web IDEà^

SAP Web IDE empowers developers to test and evaluate the functionality and performance of their apps:

Use the application preview to test functionality, design and performance before deploying it. There are different configuration options available, such as SAP UI5 runtime version, URL parameters. See [Running Applications in Development Mode](#).

Run your application using a client mock server. See [Run Applications with Mock Data](#).

Execute unit tests for SAPUI5 (Qunit) and Java (JUnit). See [Developing Application Tests and Test a Java Module](#).

Execute SAPUI5 integration tests based on OPA5. See [Performing Integration Tests](#).

URL: <https://help.sap.com/docs/btp/btp-developers-guide/programming-languages-c13bab7c106948f588e7bf5808a2ef4b>

Title: [Programming Languages | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...Programming Languages, Tools, and APIsProgramming Languages

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Programming Languages

SAP BTP supports many different programming languages; the availability of each depends on the development runtime you're using.

Environment

More Information

Cloud Foundry runtime

Development in the Cloud Foundry Environment

Kyma runtime

Development in the Kyma Environment

Related Information

SAP BTP, Serverless Runtime - Development

URL: <https://help.sap.com/docs/btp/btp-developers-guide/programming-models-programming-languages-tools-and-apis>

Title: *Programming Languages, Tools, and APIs* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...DevelopProgramming Languages, Tools, and APIs

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Programming Languages, Tools, and APIs

SAP BTP provides numerous tools for your development project when using SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime with CAP.

The following subsections give you more information about:

Programming Languages

Continuous Integration and Delivery (CI/CD)

Tools Available for SAP BTP Multi-Cloud Foundation

APIs

URL: <https://help.sap.com/docs/btp/btp-developers-guide/run-and-scale>

Title: *Run and Scale* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Run and Scale

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Run and Scale

There are two things you have to take into account to ensure your application is working properly, and bringing value:

Get constant feedback from your users and optimize the application based on this feedback.

Monitor and operate the application using the services and capabilities SAP BTP offers.

See Establish End-to-End Observability.

Security Considerations in the Run and Scale Phase

In the run and scale phase, you operate, monitor, and scale your application to ensure high availability, performance, and adaptability to growing user demands. Integrating security at this stage is essential to protect against evolving threats, maintain regulatory compliance, and support secure, scalable operations.

This phase ensures your application remains responsive and resilient as usage increases. For example, a retail company operating an e-commerce platform must defend against runtime threats such as unauthorized access or suspicious activity, while also ensuring secure auto-scaling during peak shopping periods.

By leveraging built-in security features of the SAP Cloud Application Programming Model (CAP) and ABAP Cloud, you can significantly strengthen your application defenses. Key practices include:

Continuous threat monitoring using SAP BTP observability tools to detect and respond to anomalies in real time.

Applicable for CAP: Secure auto-scaling with CAP mechanisms that dynamically manage resources while preserving tenant isolation and enforcing security policies.

Implementing these practices helps you mitigate operational risks, comply with standards like GDPR and PCI DSS, and deliver a secure, reliable user experience at scale.

The following table summarizes the security guidelines you need to be familiar with:

Secure Operations at Scale

Security Guideline

Description

Implement Continuous Threat Monitoring

Set up monitoring tools to detect and respond to runtime threats, such as unauthorized access or anomalies.
See:

Discovering Observability and Monitoring in SAP BTP, Cloud Foundry

SAP Discovery Center Mission - Implement Observability in a Full-Stack CAP Application Following SAP BTP Developer's Guide

Manage Security Patches and Updates

Regularly apply security patches to client applications and dependencies that integrate with SAP BTP to address vulnerabilities.

Secure Auto-Scaling Mechanisms

The CAP ensures secure multitenancy and supports secure auto-scaling mechanisms for multitenant SaaS applications by providing robust tenant isolation and resource management out of the box. See:

Secure Multitenancy | Capire

What Is Application Autoscaler? | SAP Help Portal

Conduct Periodic Security Audits

Perform audits to verify compliance and identify gaps in security configurations or policies.

Protect User-Generated Data Securely

Securely collect and store all forms of user-generated data—whether it originates from surveys, in-app feedback forms, comments, uploads, or integrated systems like ticketing and support platforms, to protect sensitive information and ensure consistent compliance with privacy laws across all data collection points.

Through adopting these guidelines, you significantly strengthen the security, compliance, and resilience of your SAP BTP application as it operates and scales in production. For more resources, see SAP BTP Security Recommendations | SAP Help Portal.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/starter-mission-develop-full-stack-cap-application>

Title: Starter Mission: Develop a Full-Stack CAP Application | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming ModelStarter Mission: Develop a Full-Stack CAP Application

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Starter Mission: Develop a Full-Stack CAP Application

The starter mission is called Develop a Full-Stack CAP Application Following the SAP BTP Developer's Guide and contains tutorials grouped in the following way:

The Develop a Full-Stack CAP Application group of tutorials covers the following steps for developing the Incident Management sample application:

Setting up the development environment.

Building a CAP application with SAP Fiori elements user interface and a custom logic.

Adding local launchpad, authorization, tests for local development and preparing for production.

The Deploy a Full-Stack CAP Application in SAP BTP, Cloud Foundry Runtime group of tutorials covers the following steps for deploying the Incident Management application in the SAP BTP, Cloud Foundry runtime:

Deploying the application in a productive account in SAP BTP, Cloud Foundry runtime.

Using SAP Build Work Zone, standard edition.

Setting up continuous integration and delivery pipeline in SAP BTP.

The Deploy a Full-Stack CAP Application in SAP BTP, Kyma Runtime group of tutorials covers the following steps for deploying the Incident Management application in the SAP BTP, Kyma runtime:

Deploying the application in a productive account in SAP BTP, Kyma runtime.

Using SAP Build Work Zone, standard edition.

Setting up continuous integration and delivery pipeline in SAP BTP.

See the Develop a Full-Stack CAP Application Following the SAP BTP Developer's Guide mission at SAP Discovery Center.

This image is interactive. Click the highlighted areas so you open the respective tutorial or mission.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/strengths-and-benefits-of-runtimes-and-their-programming-models>

Title: Strengths and Benefits of the Runtimes and Their Programming Models | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideStrengths and Benefits of the Runtimes and Their Programming Mode...

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Strengths and Benefits of the Runtimes and Their Programming Models

SAP BTP provides different runtimes and development models to support a wide range of application needs.

Whether you're building new cloud-native apps, extending existing SAP systems, or reusing legacy code, there's a suitable option.

SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime support modern, flexible development using Java, Node.js, and microservices, with strong support for open standards and scalability.

See SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP.

SAP BTP ABAP Environment is ideal for teams with existing ABAP skills or codebases, offering a smooth path to the cloud with strong integration into SAP S/4HANA.

See SAP BTP ABAP Environment with ABAP Cloud.

Each environment is designed to match different technical requirements, developer skill sets, and cost considerations—giving teams the flexibility to choose what fits best.

Related Informationà^

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

URL: <https://help.sap.com/docs/btp/btp-developers-guide/tools>

Title: Tools Available for SAP BTP Multi-Cloud Foundation | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...Programming Languages,

Tools, and APIsTools Available for SAP BTP Multi-Cloud Foundation

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Tools Available for SAP BTP Multi-Cloud Foundation

SAP BTP includes many tools to help you develop and manage applications. The availability of some of the tools depend on the runtime that you are running on.

Tool

Description

Account Administration in the Cockpit

The SAP BTP cockpit is the web-based administration interface of SAP BTP and provides access to a number of functions for configuring and managing applications, services, and subaccounts. Use the cockpit to manage resources, services, security, monitor application metrics, and perform actions on cloud applications.

Used for account administration in SAP BTP.

SAP Business Application Studio

Designed and optimized for business application development in SAP ecosystems, SAP Business Application Studio enhances productivity by offering specialized tools for various scenarios, including SAP Fiori application development, SAP HANA native extensions, full-stack and mobile application development, and more.

Central to the development environment is Code-OSS, the open-source foundation of Visual Studio Code, ensuring a familiar experience for developers when creating SAP-centric applications. SAP Business Application Studio streamlines the building, testing, and deployment of applications with integrated features for source control and testing. Furthermore, its Full-Stack Application Productivity Toolkit offers intuitive visual tools covering the entire development process, guaranteeing seamless integration with various SAP services and solutions.

Used for:

SAP Fiori development for ABAP and non-ABAP

CAP full-stack application development

SAP Build

SAP Build enables everyone, no matter the skill level, to create and augment enterprise applications, process automations, and business sites with drag-and-drop simplicity.

Used for low-code, no-code development.

Command Line Interface for Cloud Foundry

The Cloud Foundry command line interface enables you to work with the Cloud Foundry runtime to deploy and manage your applications.

Used for managing subaccounts in the SAP BTP, Cloud Foundry runtime.

Cloud Connector

The Cloud Connector serves as the link between on-demand applications in SAP BTP. This is the browser-based and existing on-premise systems. You can control the resources available for the cloud applications in those systems.

Used for connectivity between SAP BTP applications and on-premise systems.

Terraform provider for SAP BTP

The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP.

kubectl

The Kubernetes command line tool to communicate with a Kubernetes cluster's control plane, using the Kubernetes API.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

kubelogin

A kubectl plugin for Kubernetes OpenID Connect (OIDC) authentication.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Helm

The package manager for Kubernetes, used for installing and managing Kubernetes applications in form of Helm charts.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Paketo (Pack)

Pack is a tool maintained by the Cloud Native Buildpacks project to support the use of buildpacks. Pack lets you build container images, which are collaboratively maintained making it easier to maintain and update.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Docker Desktop

Docker Desktop is an application that enables you to manage (build, push, pull, and run) container images on your desktop and a docker-compatible command line interface.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

SAP Cloud SDK

SAP Cloud SDK provides a layer of abstractions for features of SAP BTP such as logging, multitenancy, and connectivity. It also includes project templates for different execution environments and implementations.

Used for programming in Java and JavaScript.

Service-Specific Tools

The services that run on SAP BTP can come with service-specific tools. For an overview of the services and their tools, see the SAP Discovery Center.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/tutorial-and-reference-sample-for-cross-stack-extension-application>

Title: Tutorial and Reference Sample for a Cross-Stack Extension Application | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Build and Run Multitenant SaaS Applications for Partners Who Are ...Tutorial and Reference Sample for a Cross-Stack Extension Applica...

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Tutorial and Reference Sample for a Cross-Stack Extension Application

The cross-stack application extension shows how SAP partners can build integrated solutions across SAP S/4HANA (on stack) and SAP BTP (side by side). It serves as a reference implementation for SAP partners, demonstrating how to build integrated solutions across multiple technology layers.

The cross-stack application extension enhances the Poetry Slam Manager application. See Poetry Slam Event Management on SAP BTP integrated with Sales Orders on SAP S/4HANA on GitHub.

Scenarioà

The sample scenario that the Poetry Slam Event Management on SAP BTP integrated with Sales Orders on SAP S/4HANA is about organizations that struggle to connect event planning with sales activities across

disconnected systems. Event managers lack real-time visibility into corporate sponsors for specific events, while sales teams manually coordinate between order management and event systems.

In the cross-stack application extension there are two main roles:

The event manager needs visibility which business partners sponsor each poetry slam event, requiring access to sales order details and sponsoring partner information through the Poetry Slam Manager application.

The sales representative creates sales orders in SAP S/4HANA, selects advertising products, assigns sponsoring partners, and links these commercial arrangements directly to specific poetry slam events.

The sales representative creates a sales order in SAP S/4HANA for event sponsorship. Then, she assigns poetry slam events directly from the sales order screen using automated value help. The event manager views the real-time sponsorship data in the Poetry Slam Manager application. Both the sales representative and the event manager navigate seamlessly between systems with deep-linking.

Application Architectureà

The key features of the application extension are:

Cross-stack integration: connects ABAP-based SAP S/4HANA on-stack extensions with cloud-native SAP BTP applications

Real-time data synchronization: bidirectional updates between sales orders and Poetry Slam events

In-app navigation: seamless deep linking between SAP S/4HANA and SAP BTP user interfaces

Value help integration: direct Poetry Slam event selection from sales order screens

Multitenant architecture: scalable SaaS patterns for partner solutions in Cloud ERP

The technical stack includes two sides:

On stack (SAP S/4HANA):

Remember

These cross-stack partner reference applications are delivered through multi-off delivery to enable direct and simplified consumption. This approach allows partners to quickly access the codebase, explore the solution, and experiment without additional setup.

For productive implementations of ABAP on-stack extensions within hybrid extension scenarios, the use of scalable delivery is strongly recommended.

The partner reference applications are delivered using a Z-namespace to facilitate immediate consumption. For productive deployments, the use of a registered ABAP namespace is mandatory to ensure proper ownership, prevent naming conflicts, and support compliant distribution and lifecycle management in line with scalable delivery principles.

ABAP RESTful Application Programming Model (RAP) - Core framework with CDS views, behavior definitions, validations, and custom actions

Custom data extensions: data elements, append structures, and field extensions

Cross-system communication: outbound services, communication scenarios, and service consumption model

Dynamic value help and URL navigation: seamless cross-application lookup and navigation

Lifecycle and quality: GitHub content for the cross-stack extension application is pulled in partner development system using multi-off solution, ATC checks, and change history tracking

UI adaptability: extending the user interface with custom fields

Side by side (SAP BTP)

SAP Cloud Application Programming Model (CAP)

Multitenant application architecture

Secure destination configurations

URL: <https://help.sap.com/docs/btp/btp-developers-guide/tutorials-and-reference-sample-for-full-stack-multitenant-saas-applications-for-partners>

Title: *Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Build and Run Multitenant SaaS Applications for Partners Who Are ...Tutorials and Reference Sample for Full-Stack Multitenant SaaS Ap...

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

Full-Stack Multitenant SaaS Applications for Partnersà

The Poetry Slam Manager application is a reference for SAP partners that provides you with a recommended way to becoming a SaaS provider of multitenant applications based on the SAP BTP. See Partner Reference Application 'Poetry Slam Manager' on GitHub.

The guidance covers building, running, and integrating scalable full-stack cloud applications using the SAP Cloud Application Programming Model (CAP) and adheres to the development recommendations set out in the SAP BTP Developer's Guide. It includes an ERP-agnostic design that lets you deliver your application as a side-by-side extension to consumers using any SAP solution, such as SAP S/4HANA Cloud, SAP Business One, and SAP Business ByDesign.

By using services in SAP BTP and the SAP Cloud Application Programming Model (CAP), your application meets SAP standards for enterprise-class business solutions and empowers you to focus on your application domain.

The reference application and its tutorials cover a wide range of services and components in SAP BTP. You can find a list of all services in the bill of materials of the partner reference application on GitHub. See Bill of Materials on GitHub.

The tutorials in the repository follow an incremental development approach, guiding users through the development journey. They cover:

- Developing the core application with a focus on business models and logic.

- Enhancing the application to a multi-customer (multitenant) solution.

- Integrating with various ERP backends.

- Adding more features to the application.

- Extending the application.

To cater to customer-specific and industry-specific requirements, an extensibility feature is provided as part of incremental development approach, guiding users through extension journey. See Partner Reference Application Extension.

Customer-Specific Extensionà

The Partner Reference Application Extension demonstrates how to achieve this customer flexibility by extending the Poetry Slam Manager application with enhanced catering management capabilities. By leveraging the SAP Business Technology Platform (SAP BTP) and the SAP Cloud Application Programming Model (CAP). It enables partners to build extensions that address the unique needs of individual customers while preserving the core application's functionality for other tenants and adhering to SAP standards for scalability, security, and enterprise readiness.

The Partner Reference Application Extension showcases the platform's flexibility and extensibility with features such as:

- Seamless integration with the base application

- Granular tenant-specific configurations to deliver personalized experiences

- Consistent user experience across extended and core functionality

- Secure multitenant architecture, ensuring data isolation and performance scalability

See Partner Reference Application Extension.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/tutorials-for-sap-cloud-application-programming-model>

Title: Tutorials for SAP Cloud Application Programming Model | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...Tutorials for SAP Cloud Application Programming Model

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Tutorials for SAP Cloud Application Programming Model

The Incident Management sample application is a showcase of the SAP Cloud Application Programming Model (CAP) and adheres to the development recommendations set out in the SAP BTP Developer's Guide. Best practices are documented in step-by-step tutorials based on the Incident Management application for learning how to develop, deploy, and operate a full-stack application using CAP and SAP Fiori. The tutorials also show how to use a set of important SAP BTP services in a CAP application.

When going through the sample application, you can choose from two options to try its services and runtimes before you purchase them:

Free tier for SAP BTP is best for productive projects:

Access a productive account

Start with more than 30 free-tier services and upgrade from free to paid plans

Accelerate your ramp-up with boosters and missions that include step-by-step guidance

The free-tier offering is available in the following consumption-based commercial models: the SAP BTPEA (SAP BTP Enterprise Agreement), the CPEA (Cloud Platform Enterprise Agreement) and Pay-As-You-Go for SAP BTP. See Using Free Service Plans.

SAP BTP trial is best for learning purposes:

Gain instant and full access to a trial environment

Experiment with a diverse range of more than 30 trial services for free

Get started with a variety of learning materials, such as tutorials and missions

See Start your free 90-days trial.

Before you start using the services or runtimes, you need to manage your entitlements and add quotas to your subaccounts. See Entitlements and Quotas.

Note that if you want to try out services for free, you need to select free tier service plans, if available. For a list of free services, check the Service Catalog at SAP Discovery Center.

Note

There is also a free plan for the SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime. To use the free plans for these runtimes, you have to configure the entitlements in the SAP BTP cockpit. You have one free plan for each runtime per global account and you can assign it to one subaccount at a time.

The tutorials are grouped in missions following a specific scenario. All the missions are based on the Incident Management sample application:

Starter Mission: Develop a Full-Stack CAP Application

Mission: Develop a Side-by-Side CAP-Based Extension Application

Mission: Implement Observability in a Full-Stack CAP Application

Mission: Develop an Enterprise-Grade CAP Application

Mission: Develop a Multitenant CAP Application

URL: <https://help.sap.com/docs/btp/btp-developers-guide/understanding-available-technology>

Title: Understanding Available Technology | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideUnderstanding Available Technology

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Understanding Available Technology

SAP BTP Developer's Guide has a comprehensive set of tools and services at your disposal that allow you to keep the pace and at the same time benefit from future investments in SAP BTP. The following graphic includes both ABAP and non-ABAP runtimes and provides an overview of the architecture you can use when designing and building your applications.

The overall suite of applications consists of multiple business modules that are either implemented for SAP Cloud Application Programming (CAP) or ABAP Cloud. Each business module consists of one or multiple self-contained services following a three-tier architecture with presentation, logic, and persistence layer. In both architecture styles, SAP Fiori and SAPUI5 is used to implement front end artifacts. These front end artifacts are consuming its data using OData for transactional and Information Access (InA) for analytical applications. The application logic for CAP-based applications is implemented in Node.js, Java and Typescript and is deployed in SAP BTP, Cloud Foundry runtime or SAP BTP, Kyma runtime. The application logic for ABAP Cloud is implemented in ABAP and deployed in the SAP BTP, ABAP environment. In both cases, SAP HANA Cloud is used to store relational business data.

Additional enterprise qualities are reached by integrating the business modules with complementing SAP BTP application services such as SAP Build Work Zone as a central entry point, SAP Datasphere as a cross-application data warehouse or Identity Authentication for identity authentication among others.

This image is interactive. Hover over the image and click the highlighted areas so you are placed in the respective cell in the table.

Capabilities in SAP BTP and Their Respective Services

Capability

Cloud Application Programming

ABAP Cloud

Development Tools

SAP Business Application Studio

Designed and optimized for business application development in SAP ecosystems, SAP Business Application Studio enhances productivity by offering specialized tools for various scenarios, including SAP Fiori application development, SAP HANA native extensions, full-stack and mobile application development, and more.

Central to the development environment is Code-OSS, the open-source foundation of Visual Studio Code, ensuring a familiar experience for developers when creating SAP-centric applications. SAP Business Application Studio streamlines the building, testing, and deployment of applications with integrated features for source control and testing. Furthermore, its Full-Stack Application Productivity Toolkit offers intuitive visual tools covering the entire development process, guaranteeing seamless integration with various SAP services and solutions.

ABAP Development Tools for Eclipse

SAP Business Application Studio

Use ABAP development tools for Eclipse to benefit from an efficient development environment for all ABAP-based development artifacts. Use the SAP Business Application Studio to develop the SAP Fiori parts of your ABAP-based applications.

SAP Continuous Integration and Delivery

The Continuous Integration & Continuous Delivery (CI/CD) service offers pre-configured pipelines that automate the building and testing of code changes developed in SAP Business Application Studio. These pipelines can be customized to meet your specific needs. By providing immediate feedback on the status of your code, the service helps to identify and fix issues early in the development process. This accelerates development and improves the quality of your code.

Software Delivery

SAP Continuous Integration and Delivery

By enabling deployment, setting the deployment target, and optionally configuring the handover to SAP Cloud Transport Management, you can use SAP Continuous Integration and Delivery to automate your delivery process. This helps to accelerate your delivery cycles.

ABAP Lifecycle Management

For customers:

Develop applications as software components and deliver them via the Manage Software Components application. See [Software Components](#).

A hidden Git repository is automatically managed per software component using gCTS for transport management. This process can be automated with CI/CD pipelines including steps like test automation using ABAP Test Cockpit. See [Automate the Software Lifecycle Management Process](#).

SAP Cloud Transport Management can be used to optionally integrate with higher level change management processes. See [How to Export Using SAP Cloud Transport Management](#).

For partners:

In addition to the customer scenario, products can be built based on software components with the help of the Landscape Portal to set up multitenant SaaS applications or to offer installable products like SDKs for other customers and partners. See [Landscape Portal](#).

SAP Cloud Transport Management

SAP Cloud Transport Management offers a standardized, enterprise-ready change management process. For changes in SAP BTP, Cloud Foundry runtime, the corresponding pipeline in SAP Continuous Integration and Delivery includes out-of-the-box transports in the Cloud Transport Management service.

While we strongly recommend using the Continuous Integration and Delivery service, you can also use SAP Cloud Transport Management to transport your changes without a pipeline.

Persistence

SAP HANA Cloud

Use SAP HANA Cloud service for a relational storage.

Use HANA Data Lake Files as Object Storage.

Consider compliance features like such as Audit Log, Cryptography or Customer Managed Keys that SAP HANA Cloud offers.

SAP HANA Cloud

SAP BTP, ABAP environment comes with an own ABAP-managed SAP HANA Cloud instance. Static resizing of the SAP HANA Cloud instance is supported. ABAP dictionary tables can be tagged to make use of the SAP HANA Native Storage Extensions; also, database indexes (like unique secondary indexes and fuzzy search indexes) and Dynamic View Caches can be defined. Furthermore, simple Database Partitioning based on primary keys is possible. Access to the SAP HANA instance data is only supported via the ABAP layer, for example, by using ABAP SQL, natively via ABAP-managed database procedures, and the ABAP SQL Service for external clients.

Programming Model

SAP Cloud Application Programming Model (CAP)

Use SAP Cloud Application Programming Model as programming model for non-ABAP applications.

Go-to frameworks for business application development. It supports the most widely adopted languages, which are: Java (with Spring Boot), JavaScript and TypeScript (with Node.js).

SAP Cloud Application Programming Model guarantees to run against specific versions of Node.js and Java.

Remember to plan your application to run for at least 5 years if not more.

We recommend that you choose SAP BTP services over homegrown services.

ABAP RESTful Application Programming Model (RAP)

Use ABAP RESTful Application Programming Model as a programming model within ABAP Cloud. With ABAP RESTful Application Programming Model, you can develop services for all types of SAP Fiori applications as well as publishing Web APIs.

Runtime

SAP BTP, Cloud Foundry Runtime

Use SAP BTP, Cloud Foundry runtime as a runtime for CAP-based applications.

SAP BTP, ABAP Environment

Use SAP BTP, ABAP environment for ABAP-based applications.

SAP BTP, ABAP environment delivers and enforces the ABAP Cloud development model and is based on Kubernetes. The abstraction of containers and clusters is managed by the SAP BTP, ABAP environment infrastructure.

SAP BTP, Kyma Runtime

Use SAP BTP, Kyma runtime as a runtime for CAP-based applications.

Client Library

SAP Cloud SDK

CAP is using SAP Cloud SDK behind the scenes.

SAP Cloud SDK provides client libraries for consuming OData/OpenAPI services, Destination service, and Connectivity service that extend SAP solutions and other OData/OpenAPI services on SAP Business Technology Platform. These clients allow you to concentrate on your unique business logic, simplifying development and ensuring seamless integration.

Service Consumption Model

Use the Service Consumption Model to generate local ABAP APIs to consume remote OData, SOAP and RFC services.

Connectivity

SAP Connectivity Service

Use Connectivity service to securely access on-premise systems and such hosted in virtual private cloud which are exposed via the Cloud Connector. Using the service channels feature of the Cloud Connector, it is also possible to access cloud systems such as databases from on-premise and virtual private cloud networks. For Kubernetes-based workloads, you can utilize the Connectivity Proxy to ensure secure technical connectivity from the cloud to your on-premise and virtual private cloud systems. Within the Kyma runtime, the Connectivity Proxy comes as a managed offering.

SAP Connectivity Service

The Connectivity service is automatically integrated with the BTP ABAP tenants to securely access on-premise systems and such hosted in virtual private cloud which are exposed via the Cloud Connector.

SAP Destination Service

Use Destination service to store and retrieve technical connection properties to the target systems. It automates the process of retrieving OAuth access tokens to the configured target systems. In addition, the

Destination service generates and is able to renew X.509 client certificates issued by the SAP Cloud PKI.

Communication Management

SAP BTP, ABAP Environment provides a Communication Management to integrate the custom applications with other systems to enable data exchange.

An event consumption and event provisioning is also natively supported and integrated into the ABAP RESTful Application Programming Model (RAP).

In addition, the SAP Destination Service can also be used to re-use destinations for CAP applications. See SAP Destination Service.

Transparent Proxy for Kubernetes

Use transparent proxy to enhance developers' user experience and simplify the consumption of target systems by automating the retrieval of destinations from the Destination service, as well as integration with the connectivity proxy and authentication with target systems.

Within the SAP BTP, Kyma runtime, the transparent proxy comes as a managed offering.

Application Programming Interface

SAP Business Accelerator Hub

Follow API guidelines and use the SAP Business Accelerator Hub.

User Interface (Web)

SAP Fiori Elements

SAPUI5

Use SAP Fiori elements for OData V4 to benefit from a presentation of a common UI and UX. For more flexibility, use SAP Fiori element's Flexible Programming Model with or without SAPUI5 Freestyle.

Personalization and theming are automatically supported.

SAP Fiori elements and SAPUI5 help you present one consistent solution experience to your customers, and you benefit from the upcoming designs and UX improvements automatically.

Remember that even the simplest UI components or more complex ones like geographic maps have compliance requirements like accessibility and theming.

User Interface (Mobile)

SAP Mobile Development Kit

SAP BTP SDK for Android

SAP BTP SDK for iOS

Use SAP MDK, BTP SDK for Android or BTP SDK for iOS for mobile applications.

These SDKs help you present one consistent solution experience to your customers.

Mobile application development is massively accelerated as SAP mobile services and its SDKs generate the complete synchronization and authentication layer. Additional supported features like logging, tracing, crash reporting bring extra value to you.

After deployment the Mobile services give administrators all the necessary tools and services to operate a mobile solution. Features like push, offline, mobile specific security and more simplify the operation.

Central Entry Point

SAP Build Work Zone, standard edition

SAP Build Work Zone, advanced edition

SAP HTML5 Application Repository Service for SAP BTP

Enable the central launchpad to offer a unified end-user experience.

Customers want to create a personalized view on the applications you produce. Make sure your application can be added to a central launchpad, regardless of where it runs.

SAP Fiori Launchpad for SAP BTP, ABAP Environment

SAP Build Work Zone, standard edition

SAP Build Work Zone, advanced edition

SAP HTML5 Application Repository Service for SAP BTP

SAP BTP, ABAP Environment comes with a dedicated central entry point the SAP Fiori launchpad for SAP BTP, ABAP Environment. This central entry point offers a unified end-user experience for ABAP-based applications.

It is also used as the entry point for all SAP Fiori applications to administer the ABAP system.

SAP Build Work Zone, standard edition offers a unified end-user experience across several systems for federation scenarios.

Integration

SAP Event Mesh

SAP Cloud Application Event Hub

SAP Integration Suite, advanced event mesh

SAP Event Mesh can be used to distribute events between selected SAP cloud and on-premise applications and third-party applications.

SAP Integration Suite, advanced event mesh is currently not supported by CAP and ABAP Cloud.

SAP Cloud Application Event Hub allows CAP-based applications to consume business events from supported SAP cloud applications. Currently, SAP Cloud Application Event Hub is not supported by ABAP Cloud.

SAP Master Data Integration

SAP Master Data Integration service acts as the central master data hub. It uses the integration models of SAP One Domain Model as the basis for master data replication. SAP will extend the support of SAP One Domain Model in all SAP cloud applications over time to integrate them. When out-of-the-box integration with SAP Master Data Integration is not available, SAP Integration Suite can be used to integrate with SAP ERP Central Component, SAP S/4HANA, and non-SAP applications.

SAP Integration Suite

Use the Cloud Integration capability of SAP Integration Suite for mediated data or process integration, especially if complex integration flows, transformations or dedicated protocols are required. Besides mediated application-to-application integration, Cloud Integration usage is recommended for business-to-business and business-to-governance processes.

The API Management capability of SAP Integration Suite allows you to easily enable your developer community with governed, secure, and policy-compliant access to all your APIs, events, and integrations.

Graph is a recent addition to the API Management capability of SAP Integration Suite. Graph is a powerful semantic API mediation, combining and exposing the data from diverse systems in a complex enterprise landscape as connected data graphs. The data graphs are accessed via a single, curated, and uniform data-as-a-service API and make API consumption much easier.

Cloud Integration Automation

Cloud Integration Automation service provides you a guided workflow to integrate SAP cloud solutions to on-premise and other SAP cloud solutions. The guided workflow contains instructions for manual and automated tasks to enable an easy and quick integration configuration setup.

Observability

SAP Cloud ALM

SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions.

For your CAP-based applications on SAP BTP, SAP Cloud ALM provides a central, personalized and unified operations user experience. Specifically for Java and Node.js custom-built applications in SAP BTP, Cloud Foundry runtime, there are the Data Collection Instrumentation Libraries based on Open Telemetry designed to enable the data collection infrastructure in SAP Cloud ALM. With these libraries, you can collect data for many observability use cases, such as:

Real User Monitoring

Health Monitoring

Integration and Exception Monitoring

SAP Cloud ALM

All ABAP cloud applications are supported by SAP Cloud ALM for central observability. SAP Cloud ALM provides the following use cases:

Real User Monitoring

Health Monitoring

Synthetic User Monitoring

Integration Monitoring

Job and Automation Monitoring

Data Forwarding to SAP Focused RUN

For Health Monitoring, you could extend the delivered content with your own custom metrics.

SAP BTP, ABAP Environment strictly distinguishes between platform monitoring and application monitoring.

The platform monitoring, like availability monitoring, is ensured by the service itself.

For application monitoring, which is in your responsibility, respective tools are offered. Use the Technical Monitoring Cockpit to analyze and optimize the application on-stack:

System workload

Resource consumption and capacity

Detailed statistics captured for single requests

Outbound communication

SQL statements

SAP Alert Notification service for SAP BTP

Local expert tool that allows you to subscribe to events from the platform – such as from used services, from your custom-built apps, or from the infrastructure – and to consume them via your channel of choice (such as by receiving notifications via email or in your preferred chat application). Can be integrated into central alerting of SAP Cloud ALM.

SAP Cloud Logging

SAP Cloud Logging allows you to analyze your SAP BTP workloads in great detail regarding performance, errors, usage, and other characteristics.

It covers processing of logs, metrics, traces across SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime with flexible storage, alerting, and dashboarding.

Customer Landscape Discovery

Unified Customer Landscape

Use Unified Customer Landscape service for customer landscape management.

There are different ways to add systems to the System Landscape page in the SAP BTP cockpit: manually or automatically. If a system of your solution is associated with your global account or through a subscription in SAP BTP cockpit associated with a given subaccount, it will appear in the list automatically. Otherwise, you have to add your system manually. Systems are added to the list in one of the following ways:

Auto-Discovered

An auto-discovered system is a system (associated with the given global account) that has been discovered and added automatically to the list based on information of the existing system landscape. Any SAP system of the supported system types that is associated with the same customer ID, with which your global account in SAP BTP is associated, will be added automatically in the system landscape list.

Subaccount/<my-subaccount>

Specifies that the system has been added through a subscription in SAP BTP cockpit associated with a given subaccount. The subscription has been discovered and added automatically through the subaccount.

Manually-Added

Specifies that the system has been added to the list manually by the global account administrator, using the Add System button and completing the wizard. The system has been associated with the global account in SAP BTP.

Landscape Portal

Landscape Portal is offered to manage all the systems within your global account in SAP BTP. It shows the list of the ABAP systems, it enables the control of system hibernation to reduce costs and it offers features to control the system lifecycle, for example, to nominate systems for the pre-upgrade prior to the standard upgrade of a quarterly ABAP platform release.

Extensibility and Integration

Side-by-Side Extensibility with Unified Customer Landscape:

Register Systems

Create Formations

SAP SuccessFactors Extensibility Service

SAP S/4HANA Cloud Extensibility Service

Use Unified Customer Landscape service that offers customer landscape management capabilities for your SAP S/4HANA, SAP Ariba, SAP SuccessFactors, and other SAP and third-party systems in one single experience.

In the SAP BTP cockpit, you get a comprehensive overview of all your systems associated with your customer ID. These systems can be registered or auto discovered. They are conveniently listed as a record in the Systems list in the System Landscape page in the SAP BTP cockpit. Moreover, Unified Customer Landscape lets you integrate one or more systems in a common business case by including these systems in a formation.

On-Stack Extensibility by Extending SaaS Applications

In addition to the standard side-by-side approach for core applications such as SAP S/4HANA, SAP BTP, ABAP

Environment also offers two extensibility options within the product itself:

With developer extensibility, you can extend partner solutions which are installed in your customer system, for example by adding custom fields, custom nodes and business logic.

With key user extensibility, you can extend a multitenancy SaaS application offered by a partner. The supported extensibility features to extend SaaS applications are UI adaptations, custom fields and custom logic (implement Business Add-Ins).

You can create business configuration objects. See Business Configuration.

You can create and set up integration scenarios by maintaining communication arrangements.

Data Privacy

SAP Data Privacy Integration

Use Data Privacy Integration service to make your SAP BTP application compliant with the corporate Data Privacy and Protection standards. Integrate with the Data Privacy Integration service to support cross consumable Data Privacy and Protection features for our customers and support cross Data Privacy and Protection compliance in end-to-end processes.

Workflow

SAP Task Center

SAP Task Center service enables integration with SAP applications to provide a single entry point for end users to access all their assigned tasks. The tasks can be accessed by end users through the SAP Task Center Web application. SAP Task Center helps you integrate tasks into a central solution.

Use SAP Task Center as a unified inbox for tasks across multiple applications with integrated user experience. Tasks from multiple SAP solutions are gathered in one list and ready to be processed in just one click, shortening the completion time for business-critical tasks. For example, business users can process all their tasks from the connected systems, without the need to switch and log in separately into different inboxes.

SAP Build Process Automation

SAP Build Process Automation provides a simpler and faster way to adapt, improve, and innovate business processes with drag-and-drop simplicity.

The solution combines workflow management, SAP Intelligent Robotic Process Automation (SAP Intelligent RPA) functionality, decision management, process visibility, and embedded AI capabilities into one intuitive low-code experience.

You can start automation projects with hundreds of process content packages, SAP Intelligent RPA bots and connectors designed specifically to enhance the capabilities of the SAP solutions you are using. It provides a unified development experience for users of all skill levels enabling fusion teams of business experts and developers to work together and solve challenges faster.

Job Scheduling

SAP Job Scheduling Service

SAP Job Scheduling service allows you to define and manage jobs that run once or on a recurring schedule.

Use this runtime-agnostic service to schedule action endpoints in your application or long-running processes using Cloud Foundry tasks. Use REST APIs to schedule jobs, including long-running jobs asynchronously, and create multiple schedule formats for simple and complex recurring schedules. Manage jobs and tasks and manage schedules with a web-based user interface.

Application Jobs

Job scheduling is integrated into the product. Application Jobs can be defined, implemented and monitored.

Application Logs

Application Logs

SAP Task Center helps you integrate tasks into a central Application Logging is integrated into the product.

Application Logs can be defined, implemented, and monitored.

Analytics

SAP Analytics Cloud

Use embedded analytics scenarios in your application including dashboards, multi-dimensional reports and KPIs.

SAP Analytics Cloud

Use SAP Analytics Cloud on top of Information Access (InA)-enabled Core Data Service analytical models. Furthermore, Dragonfly-based multi-dimensional reporting can be used to integrate Embedded Analytics functionality in a component-based and SAP Fiori-native way.

SAP Datasphere

For analytics across applications use SAP Datasphere.

SAP Datasphere

Via the ABAP SQL Service it is possible to integrate with SAP Datasphere. Furthermore, the ABAP SQL Service in combination with ABAP ODBC Driver or the SAP HANA Cloud ABAP SDA Adapter allows data federation via external clients respectively via SAP HANA Cloud-based SAP BTP applications.

Document Management

SAP Document Management Service

Document Management Service, Integration Option lets you build document management capabilities for your business applications using the integration component or the easy-to-use, reusable UI component.

Document Management Service, Application Option is a standalone, ready-to-use web application that provides document management capabilities for your enterprise content.

Object Store on SAP BTP

Object Store service on SAP BTP lets you store and manage objects, which involves creation, upload, download, and deletion. This service is specific to the Infrastructure-as-a-Service layer such as Azure Blob Storage, Amazon Web Services, and Google Cloud Platform.

Low Code/No Code

SAP Build

SAP Build is a low-code offering to accelerate development and automation. It enables you to create enterprise apps (SAP Build Apps), automate processes (SAP Build Process Automation), and design business sites (SAP Build Work Zone) without writing code. Accelerate development with prebuilt connectors and business content for SAP and third-party systems to integrate seamlessly. SAP Build fosters collaboration between business and development teams with built-in governance and lifecycle management.

Service Management

SAP Service Manager

SAP Service Manager service allows you to consume platform services in any connected runtime environment, track the creation and management of service instances, and share services and service instances between different environments.

Authentication

SAP Authentication and Trust Management Service

The SAP Authentication and Trust Management service lets you manage user authorizations and trust to identity providers. Identity providers are the user base for applications.

We recommend that you use an Identity Authentication tenant, an SAP on-premise system, or a custom corporate identity provider. User authorizations are managed using technical roles at the application level, which can be aggregated into business-level role collections for large-scale cloud scenarios.

Access Management

User authorizations are managed and can be aggregated into business roles locally in SAP BTP, ABAP environment.

Identity Authentication

The Identity Authentication service provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options.

Identity Authentication

The Identity Authentication service provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options.

Technical users for system-to-system communication are managed locally in the SAP BTP, ABAP environment. SAP BTP, ABAP environment supports mTLS and basic authentication as authentication options for technical users.

Audit Logging

SAP Audit Log Service

SAP Audit Log is a core, security, and compliance-based SAP BTP service to provide means for audit purposes. The default and advanced capabilities of Audit Log Service are available for SAP BTP applications and services.

Manage Security Audit Log

Security audit logging is provided automatically by the SAP BTP, ABAP Environment. It can be configured by the Manage Security Audit Log administration application.

Read Access Logging

Read Access Logging (RAL) is used to monitor and log read access to sensitive data. This data may be categorized as sensitive by law, by external company policy, or by internal company policy.

Security

SAP Credential Store

SAP Credential Store service provides a repository for passwords, keys and keyrings for applications that are running on SAP BTP. It enables the applications to retrieve credentials and use them for authentication to external services, or to perform cryptographic operations and TLS communication. SAP Credential Store is exposed to the applications via a REST API.

Communication Management

The ABAP environment offers its Communication Management. It contains a credentials store which allows the applications to perform outbound communication using the credentials for authentication to external services and TLS communication.

Identity Management

Identity Provisioning

The Identity Provisioning service automates identity lifecycle processes. It helps you provision identities and their authorizations to various cloud and on-premise business applications.

Identity Provisioning

The Identity Provisioning service automates identity lifecycle processes. SAP BTP, ABAP environment supports the Identity provisioning service to provision business users and their assignment to business roles.

Technical Aç 2 utomation

Terraform Provider for SAP BTP

The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP.

SAP Automation Pilot

SAP Automation Pilot allows to reduce operational efforts to run your cloud applications on SAP BTP by automating them. For this, it brings catalogs of automated commands, such as around database and application lifecycle management tasks for the different runtimes of SAP BTP. Use these commands to model and run your custom scenarios, optionally combined with custom scripts. Also, you can describe your needs to let the included Generative AI feature come up with a command for you. The created commands can be triggered manually, via scheduling or externally, such as from SAP Cloud ALM.

Feature Toggling and Delivery

SAP Feature Flags

SAP Feature Flags service is designed to help you manage and control the deployment of new features within your applications. You can activate and deactivate features dynamically without the need for redeployment, enabling a more flexible and responsive development process.

SAP Feature Flags service supports gradual feature rollouts, allowing you and your team to test new functionalities with specific user groups before a full-scale launch. Additionally, it provides the capability to quickly disable features if issues arise, minimizing potential disruptions. The service integrates seamlessly with existing SAP environments, ensuring that feature management is consistent and efficient across different applications.

Overall, SAP Feature Flags Service offers a practical solution for managing feature releases, enhancing the ability to deliver stable and reliable software updates.

Related Informationà^

UX Design and Product Compliance Standards

Introducing Continuous Integration and Delivery (CI/CD)

URL: <https://help.sap.com/docs/btp/btp-developers-guide/user-experience-design>

Title: User Experience Design | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideDeveloping with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Ky...DesignUser Experience Design
SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

User Experience Design

Once you have a sufficient understanding of your users' needs, by successfully completing the Discover phase, you can start designing the user experience of your application.

User experience goes beyond the user interface, and covers all aspects of the interactions a user has with the application: getting the right information shown at the right time, allowing the user to get their work done with only a few clicks needed, providing a fast and accessible solution, supporting all the international languages your users use in their systems.

A product with good usability gives users a good user experience.

Usability is defined by the degree to which these aspects are provided to a user to get their tasks or jobs done:

Effectiveness

Efficiency

Ease of learning (Intuitiveness)

Error tolerance

Satisfaction

Consistency contributes to ease of learning and efficiency.

When providing users with applications to complement or extend your SAP landscape, you will need to ensure that your applications have a similar look and feel to SAP's standard applications, and hence provide a consistent user experience for your users. This helps to make the applications intuitive to use (ease of learning) and helps with efficiency, since they look and behave the same as other applications users are already familiar with. For this reason, we recommend that you follow the SAP Fiori for Web Design Guidelines. In particular, follow the Best Practices for Designing SAP Fiori Apps.

In addition to consistency, ensure that your application is accessible, by following the guidance provided by Accessibility in SAP Fiori. Also, be sure to provide language support for all relevant languages as well as providing suitable user assistance.

When designing your application, make sure that you continue with your user research, so that you can get quick feedback about design ideas and artifacts. Iterate often, with the motto "Fail fast, fail early" – in other words, sketch out solutions, get feedback, try again, and keep on iterating until you are confident that your design is what your users need.

For more information about user research during the design phase, have a look at the User Research Resources related to the Design phase. The methods outlined there contain the most commonly practiced user research methods at SAP.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/using-multitarget-applications-to-manage-dependencies>

Title: *Using Multitarget Applications to Manage Dependencies | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's Guide...DevelopUsing Multitarget Applications to Manage Dependencies

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

Using Multitarget Applications to Manage Dependencies

One challenge of moving into the cloud is deploying applications that consist of multiple interconnected components.

In Cloud Foundry, we recommend that you develop multitarget applications that let you package those components into one bundle, and deploy and manage them all at once. Cloud applications often come with a lot of heterogeneity, which is one of the key strengths of cloud development, allowing for agility, resilience, and the rapid development of new features. However, it also increases the complexity of cloud applications, which:

Usually consist of multiple interdependent software modules

Are written in different programming languages using multiple development tools

Might involve different products

May be deployed to multiple target runtimes

A combined lifecycle lets you deploy all parts together, automatically, and in the right order, and manage the configuration of the complete solution. You can achieve such a combined lifecycle by developing multitarget applications. Each multitarget application has the following characteristics:

One archive file that includes all modules and a description of the dependencies

Can be delivered, transported, linked to SAP software components, and deployed

The process can be automated in a continuous integration pipeline

The multitarget application archive contains all required application types and configurations, as well as a deployment descriptor file. It is intended to be used as a generic artifact that can be deployed and managed on several SAP BTP subaccounts. For example, you can use one multitarget application archive in your development subaccount and reuse it in your production subaccounts.

As all interdependencies are part of the archive file, it's easy to pass multitarget applications from development to operations. All required information for deployment is provided during the development process. Due to the benefits provided by applying the multitarget application approach, it is also part of the SAP Cloud Application Programming Model.

Recommendation

The approach isn't mandatory for applications that are running on SAP BTP – you can also develop without applying it. Without the multitarget application approach, you'll need to manually deploy your application artifacts, for example by triggering the deployment from SAP Business Application Studio or manually uploading artifacts via SAP BTP cockpit.

We recommend that you use multitarget applications in the following cases:

You're developing a business application composed of several different parts – apps, services, content, and others – that you want to manage as a single unit.

Your business application has dependencies to external resources, such as backing services (database, messaging, and so on), APIs, and configurations from other applications.

Your business application has a certain default configuration, for example memory, disk, number of individual app instances, environment variables, service plans, and others.

For more conceptual information about multitarget applications and detailed step-by-step instructions, see Multitarget Applications in the Cloud Foundry Environment.

There are several options to create multitarget application archives:

If you use SAP Business Application Studio, see MTA Development.

If you use SAP Web IDE Full-Stack, you can use multitarget application templates for Cloud Foundry applications, where the descriptor file is maintained automatically, for example, whenever you add a new module in the SAP Web IDE.

If you have development modules from other sources, you can use the multitarget application archive builder, a Java-based command-line tool that builds modules and packages them into a deployable multitarget application archive, together with a deployment descriptor. It is available for download from SAP Development Tools (see <https://tools.hana.ondemand.com/#cloud>).

If you use Continuous Integration and Delivery (CI/CD), you can let the pipeline perform an automated build of your changes pushed to your central source code repository. The outcome can also be an MTA archive. For more information, see Delivering Applications.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/ux-design-and-product-compliance-standards>

Title: UX Design and Product Compliance Standards | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Developer's GuideUnderstanding Available TechnologyUX Design and Product Compliance Standards

SAP BTP Developer's GuideEnglish

Provide feedback on our search

à

UX Design and Product Compliance Standards

Applications built on SAP BTP need to provide a consistent and compliant user experience across runtimes, while giving developers flexibility to address diverse business requirements. The SAP Fiori design language is the standard for achieving this goal.

From a development perspective, there are three main recommendations for creating user interfaces:

SAP Fiori elements

Flexible Programming Model

Freestyle SAPUI5

SAP Fiori Elementsà

SAP Fiori elements is a framework on top of SAPUI5 that accelerates the development of SAP Fiori

applications by providing predefined templates and patterns based on OData services, ensuring consistent UI design and functionality without extensive coding. Provides the fastest development speed, lowest total cost of ownership, and out-of-the-box compliance with SAP Fiori design guidelines.

Floorplans

Floorplans in SAP Fiori elements are standardized UX design templates that guide the layout and behavior of applications. They include variations like List Report, Object Page, and Overview Page, each serving specific business needs and ensuring a cohesive user experience across SAP applications.

Building Blocks

Building blocks in SAP Fiori Elements are reusable UI components that encapsulate common functionalities, such as data visualization or form entry. They streamline development by enabling consistent application behavior and enhancing modular design within SAP Fiori applications.

Flexible Programming Model[^]

A hybrid approach that extends SAP Fiori elements with custom UI logic where needed. Recommended for applications that follow standard floorplans but require selective flexibility (for example, custom actions, advanced visualizations). Balances reuse of compliant templates with targeted freedom for developers.

Extensions

Standard templates offer multiple extension points, allowing you to enhance them with custom elements.

Custom Pages

When developing custom pages, leveraging the SAP Fiori elements base component enables the use of building blocks and additional features like routing, all readily accessible.

Freestyle SAPUI5[^]

Full control over UI logic and layout using SAPUI5 as a UI framework. Recommended only when business scenarios demand highly customized interfaces that cannot be covered by SAP Fiori elements or the flexible programming model. Requires higher development and maintenance effort but still benefits from the Horizon theme and SAP Fiori guidelines.

SAP Fiori Tools

SAP Fiori tools provide many capabilities to increase the efficiency of developing SAP Fiori applications using either SAP Fiori elements or the freestyle SAPUI5 approach. SAP Fiori tools, together with SAP Fiori elements, reduce development time, maintenance costs, and leverage the advantages of a metadata-driven UI. SAP Fiori tools include the following:

- Application Wizard for the initial creation of an application.

- Service modeler for viewing the data model.

- (SAP Fiori elements only) XML and form-based editor for maintaining of annotations.

- (SAP Fiori elements only) Application page structure and ability to configure the SAPUI5 flexibility settings.

- (SAP Fiori elements only) Guided development for implementing features.

Compliance Standard[^]

Applications designed with the recommended UX approaches automatically inherit a broad set of SAP product compliance standards. These ensure that applications are enterprise-ready without requiring developers to implement them from scratch.

Out-of-the-Box Compliance

By using SAP Fiori elements or the flexible programming model on top of SAPUI5 with Horizon theme, applications automatically comply with:

- Accessibility (WCAG 2.2) – support for screen readers, ARIA roles, keyboard navigation.

- SAP Fiori design guidelines – consistent floorplans, navigation, and semantics.

- Responsive design – adaptive layouts for desktop, tablet, and mobile.

- Performance best practices – SAPUI5 optimizations for data loading and rendering.

- Security standards – built-in protection (XSRF, CSRF tokens), secure authentication with Identity Authentication service/SAP Authorization and Trust Management service.

To be fully compliant, applications must also:

- Expose OData V4 services (CAP, ABAP Cloud, or Node.js) with proper annotations.

- Enable internationalization (i18n) for texts and labels.

- Respect SAP Fiori design guidelines when customizing UI (colors, icons, navigation).

SAP BTP Developer's Guide

The SAP BTP Developer's Guide is the starting point for developing a business application on SAP BTP. It contains recommendations and best practices that give you an overview of what you should consider when working on development projects on SAP BTP. It also contains links to step-by-step instructions when required.

Is This Guide for You?à^

If you're an architect or a development project lead, this guide helps you to plan the architecture of your application and the services on SAP BTP you can use to achieve your goal.

If you're a developer, this guide helps you to define the correct methodologies and tools for your development project.

If you're an SAP partner, this guide helps you to set up SAP BTP for developing and running applications and services for your customers. See [Build and Run Multitenant SaaS Applications for Partners Who Are Independent Software Vendors](#).

Note

If you're an administrator, you have to set up the correct organizational structure to create an account and security model. In this case, use the [SAP BTP Administrator's Guide](#).

Note

This guide is targeted at customers who want to run and use applications in a production environment. If you're an SAP BTP trial user, you might still find that some information in this guide is useful. Check out the following page for more details about trial accounts: [Trial Accounts and Free Tier](#). Please note that the services available in the trial version differ from the ones in the enterprise version.

Prerequisitesà^

If you're new to SAP BTP, read:

SAP Business Technology Platform

SAP BTP is a cloud platform that brings together application development, automation, data management, analytics and planning, integration, and AI capabilities into one unified environment optimized for SAP applications. It offers:

An intuitive, modern development environment for both, professional IT and citizen and business developers, Built-in data models, integrations, workflows, application templates, and AI business services

Self-service data discovery, modeling, planning, and analytics for business users in a governed environment

SAP BTP Adoption Framework

This is the central access point for architects, developers, and administrators to build and run enterprise-grade solutions on SAP BTP. It comprises decision guides, reference architectures, methodologies, recommendations, and DevOps principles.

Basic Platform Concepts

Contains regions, runtimes, accounts, members, quotas, capabilities and links to in-depth explanations.

Getting Started Checklist

Explains your responsibilities and SAP's responsibilities when it comes to application lifecycle management.

SAP BTP Administrator's Guide

Helps you plan and set up your landscape and your lifecycle management for running applications on SAP Business Technology Platform (SAP BTP). It's part of the SAP BTP Adoption Framework and contains recommendations for planning development projects – from setting up the correct organizational structure to creating an account and security model, to operating applications.

SAP Discovery Center

Helps you turn your data into business value with SAP BTP. In the SAP Discovery Center you can find end-to-end scenarios, called missions, the Service Catalog of SAP BTP, and reference architectures.

SAP Architecture Center

Provides solution reference architectures, helping businesses adopt SAP solutions to turn data into valuable business insights.

How to Use This Guideà^

Use the SAP BTP Developer's Guide to help you implement business applications on SAP BTP. This guide is

part of the SAP BTP Adoption Framework and explains the building blocks for developing, delivering, and integrating business applications.

Understand the Available Technology

Understanding the development use cases, recommended services, and the continuous integration and continuous delivery (CI/CD) principles offered by SAP Business Technology Platform (BTP) is essential to streamline application development, deployment and integration within the SAP ecosystem.

See [Understanding Available Technology](#).

Select Runtime and Programming Model

Choosing a suitable runtime and the corresponding programming model is crucial to optimize application development, deployment and management on SAP BTP. There are different strengths and benefits you can explore before deciding:

[SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes with CAP](#)

[SAP BTP ABAP Environment with ABAP Cloud](#)

Go Through the Development Phases

Development projects for business applications have similar characteristics. Standardized development guidance is driving developer efficiency.

Based on the experiences of successful business applications, this guide condenses best practices and technologies that can be safely recommended to you. These recommendations are structured following the design-led development phases:

The focus of the SAP BTP Developer's Guide is on the design, deliver and run and scale phases. However, you can also find information related to the explore and discover phases.

Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It supports Java (with Spring Boot), JavaScript, and TypeScript (with Node.js), which are some of the most widely adopted languages. CAP guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

Explore

Identify the business opportunity and set up the roles and responsibilities in your team.

Discover

Design

User Experience Design

Design the user experience of your application to provide good usability.

Technology Design

Use domain-driven design to create an application that is modularized and has shorter innovation cycles and a clearer focus.

Deliver

Set Up

Before you begin developing your applications, make sure your landscape setup is appropriate for managing their lifecycles.

Develop

Develop your application by following the recommendations outlined in this section about the two programming models, the numerous programming languages, the respective development tools and also about aspects such as multitenancy, security, extensibility and others.

Run and Scale

Get constant feedback from your users, optimize the application based on this feedback, and operate the application using the services and capabilities SAP BTP offers.

Developing with ABAP Cloud in the SAP BTP ABAP Environment

ABAP Cloud reflects the modern way to develop ABAP. It allows you to build lifecycle-stable and cloud-ready business applications, services, and extensions.

Contribute to the SAP BTP Developer's Guide

Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding

option on SAP Help Portal:

Feedback Create issue: Provide feedback about a documentation page. This option opens an issue on GitHub.

Feedback Edit page: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

Contribution Guidelines

Introduction Video

Introduction Blog Post

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/11fbee6f83ad4a70819775f72b65773c.html>

Title: Extensions, Neo Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP, Neo EnvironmentExtensions, Neo Environment

SAP BTP, Neo EnvironmentEnglish

Provide feedback on our search

à

Extensions, Neo Environment

The extension capabilities of SAP Business Technology Platform (SAP BTP) enables developers to implement loosely coupled extension applications securely, thus implementing additional workflows or modules on top of the existing SAP cloud solution they already have.

Caution

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

Tip

This documentation refers to SAP Business Technology Platform, Neo environment. If you are looking for documentation about other environments, see SAP Business Technology Platform .

Introductionà

All standard SAP solutions are offered with customizing capabilities. Additionally, customers often have their own requirements for innovative or industry-specific extensions and the extension capability of SAP BTP can help them build, deploy, and operate their new functionalities easily and securely.

Extension Applicationà

SAP BTP, Neo environment offers ready-to-use development and runtime environment in the cloud. You can extend standard SAP solutions without disrupting their performance and core processes. When building extension applications, you can also benefit from the automation of the integration between the SAP BTP, Neo environment and the extended SAP solutions.

These are the specifics of an extension application:

The extension application needs to call APIs from the SAP solution.

The SAP solution (for example, SAP SuccessFactors) can notify the extension application using APIs exposed from the extension application.

The UI of the extension application can:

Be embedded in the UI of the SAP solution

Be separate from the SAP solution, for example embedded into a corporate or a consumer portal

Extension Subaccountà

An extension subaccount is part of a customer or partner global account in SAP BTP which is configured to interact with a particular SAP solution through standardized destinations, usually with identity propagation turned on. This subaccount is paired with the extended SAP solution.

Provide Extension Applicationsà

There are two deployment models for providing an extension application:

Provider-managed deployment model: an extension application is deployed in a separate subaccount in SAP BTP, and the customer's extension subaccount is subscribed for this application.

For example, an SAP partner can use this deployment model to offer his extension applications as a software-as-a-service (SaaS).

Consumer-managed deployment model: an extension application is deployed directly in the extension subaccount of the customer.

This deployment model can be used for extension applications built by the customer, or purchased from an

SAP partner as a deployable solution.

Related Information

Extending SAP SuccessFactors in the Neo Environment

Extending SAP S/4HANA Cloud in the Neo Environment

Extending SAP Cloud for Customer in the Neo Environment

Extending SAP Ariba in the Neo Environment

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/34ac79024d41469a804ba9b9deeb5b1d.html>

Title: What Is SAP BTP, Neo Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP, Neo Environment

SAP BTP, Neo EnvironmentEnglish

Provide feedback on our search

à

What Is SAP BTP, Neo Environment

SAP BTP, Neo environment is an enterprise platform-as-a-service (enterprise PaaS) that provides comprehensive application development services and capabilities, which lets you build, extend, and integrate business applications in the cloud.

Caution

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

Tip

This documentation refers to SAP Business Technology Platform, Neo environment. If you are looking for documentation about other environments, see SAP Business Technology Platform .

At its core, SAP BTP enables you to do the following:

Extend on-premise and cloud solutions

Personalize and extend your existing applications, regardless of whether they run on-premise or in the cloud, to achieve agility and go to market faster.

Integrate on-premise and cloud solutions

Bring together disparate systems and IT landscapes to streamline your business processes and increase efficiency.

Develop new cloud applications

Create new cloud applications with a small capital investment in software and hardware.

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

The Neo environment also allows you to deploy solutions on SAP BTP. In the context of SAP BTP, a solution is made up of various application types and configurations created with different technologies, designed to implement a certain scenario or task flow. You can deploy solutions by using the Change and Transport System (CTS+) tool, the console client, or the SAP BTP cockpit, which also lets you monitor your solutions. The SAP multitarget application (MTA) model encompasses and describes application modules, dependencies, and interfaces in an approach that facilitates validation, orchestration, maintenance, and automation of the application throughout its life cycle.

The Neo environment lets you use virtual machines, allowing you to install and maintain your own applications in scenarios that aren't covered by the platform.

Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

Feedback Create issue: Provide feedback about a documentation page. This option opens an issue on GitHub.

Feedback Edit page: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

Contribution Guidelines

Introduction Video

Introduction Blog Post

Capabilities and Services

According to your use cases, you may want to consume a set of services that are provided by SAP BTP. For more information, see [Solutions and Services](#).

Regions

You can deploy applications developed in the Neo environment to various SAP data centers around the world. For more information about regional availability of the Neo environment, see [Regions and Hosts Available for the Neo Environment](#).

Integration with SAP and Non-SAP Software

SAP BTP facilitates secure integration with on-premise systems that are running software from SAP and other vendors. Using platform services such as the Connectivity service applications can establish secure connections to on-premise solutions, enabling integration scenarios with your cloud-based applications. For more information about the Connectivity service, see [What Is Connectivity for the Neo Environment?](#).

Secure Data

The comprehensive, multilevel security measures that are built into SAP BTP are engineered to protect your mission-critical business data and assets, and to provide the necessary industry-standard compliance certifications.

Quality Certificates

Third-party certification bodies provide independent confirmation that SAP meets the requirements of international standards. You can find all certificates at <https://www.sap.com/corporate/en/company/quality.html>.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/btp-development>

Title: *Development | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformDevelopment

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Development

Develop and run business applications on SAP Business Technology Platform (SAP BTP) using our cloud application programming model, APIs, services, tools, and capabilities.

SAP BTP supports multiple environments. It features a number of tools and programming languages that increase your flexibility and freedom when developing applications. You can choose the environment that best suits your use cases and skill set and get to work. You also have the possibility to integrate applications you've built with other solutions by SAP.

Environment options

Cloud Foundry

Kyma

ABAP

Benefits

Simplified developer experience for business application development

Large choice of programming languages

Intuitive “code-to-container” packaging and deployment, managed by the platform

Platform-managed application security patching and updates

Automatic application routing, load balancing, health checks, and multilevel self-healing

Support for CAP – an opinionated business app development framework

Take full advantage of the advanced features and rich ecosystem of Kubernetes

Free choice of programming languages and models (containerized deployments)

Combines microservices and serverless functions

Built-in, managed service mesh based on Istio, and other cloud-native open-source modules to reduce the development effort

Built-in, managed event mesh

Managed infrastructure: day-2 operations, security patches, and updates

Full administrator access

Refined horizontal and vertical automatic scalability

Dedicated application runtime

Zero downtime infrastructure setup by default

Support for CAP – an opinionated business app development framework

Support for on-premise connectivity

ABAP programming language

Fast prototyping with ABAP RESTful Programming Model (RAP)

Integrated development lifecycle

Reuse existing on-prem ABAP assets

Additional Information

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

Development in the ABAP Environment

Shared Benefits

No infrastructure vendor lock-in

Build scalable multitenancy business applications (SaaS)

Out-of-the-box consumption of SAP and hyperscaler services

Built on industry standards and open technology

Good For

Managed build-on approach

Enterprise-grade business applications and services

Cloud-native web applications and services

Scalable, microservice-based applications

Small to medium extensions built with CAP/low-code tooling

Open build-on approach

Enterprise-grade applications

Cloud-native development of apps and services

Low latency infra-services communication

Reduced infrastructure management effort

Highly scalable, microservice-based applications

Applications built with CAP

User-centric process extensions

Robust, transactional cloud applications

Migrating and adapting add-ons to the cloud

Reusing existing on-premise ABAP code

Enabling ABAP developers to go to the cloud

Skills

Any major programming languages

SAP Fiori/UI5 and SAP HANA

Kubernetes knowledge

Docker

NodeJS or Python for serverless functions

Any major programming language

SAP Fiori/UI5 and SAP HANA

Ability to write modern ABAP code

Core data services

SAP Fiori/UI5 and SAP HANA

We provide detailed information about developing, configuring, and deploying your applications depending on your preferred environment and development approach.

Development in the Cloud Foundry Environment

Learn more about developing applications on the SAP BTP, Cloud Foundry environment.

Development in the ABAP Environment

Learn more about developing applications in the ABAP environment.

Development in the Kyma Environment

Learn more about developing applications in SAP BTP, Kyma runtime.

Related Information

SAP BTP Administrator's Guide

Developing with the SAP Cloud Application Programming Model

Consuming Services in SAP BTP

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/cloud-application-programming-model>

Title: SAP Help Portal | SAP Online Help

ProductsWhat's NewExplore SAP

Loading Document...

âƒ

â„

âĶ

à PDF

àĭFavorite

á=Share

áÀ

áŒTable of Contents

Loading Table of Contents...

Loading Topic...

âˆWas this page helpful?

â"\"Yes

â#No

Copyright

Disclaimer

Privacy Statement

Legal Disclosure

Trademark

Terms of Use

Cookie Preferences

Accessibility & Sustainability

Ask a Question about the SAP Help Portal

Find us on

Share

á=

Your choices regarding cookies on this site

This site uses cookies and related technologies, as described in our Cookie Statement, for purposes that may include site operation, analytics, enhanced user experience, or advertising. You may choose to consent to our use of these technologies, or manage your own preferences.

Privacy Statement | Cookie Statement

Accept All

Deny All

Manage Settings

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/creating-employee-record-for-new-developer>

Title: Creating an Employee Record for a New Developer | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Creation of

Developer UsersCreating an Employee Record for a New Developer

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Creating an Employee Record for a New Developer

Create an employee record for a new developer user for the ABAP environment.

Procedure

Log on to the administration launchpad of the ABAP environment as administrator (see Logging on to the Administration Launchpad of the ABAP Environment).

Choose the Maintain Employees tile.

Choose Create.

Enter the data for the new developer, including employee ID, name, and e-mail address.

The entered e-mail is important because it represents the mapping between the user in the identity provider and in the ABAP environment.

Choose Save.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/developing-user-interface>

Title: *Developing User Interface | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology

PlatformDevelopmentDevelopment in the Cloud Foundry EnvironmentDeveloping User Interface

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Developing User Interface

Get information on user interface development in the SAP BTP, Cloud Foundry environment.

The key to a modern user interface is pairing great performance with a consistent and recognizable design across applications, platforms, and devices. To keep the user experience homogenous, you can establish a flexible implementation paradigm based on reusable components. Doing so enables extensibility and easier maintainability.

We know that it can be difficult to decide this up front. You can find information about UI development options in the SAP BTP, Cloud Foundry environment in the following sections.

Fiori Design Guidelinesà^

The SAP Fiori Design Guidelines define design solutions that are easy to use, robust, scalable, and of enterprise-grade quality. The guidelines are independent of the implementation technology.

The modular design approach focuses on user tasks and workflows. It enables seamless transitions between applications and offers clear guidance for extension and integration design. To ensure a consistent user experience, we recommend following the Fiori Design Guidelines independent of the UI technology you plan on using.

SAPUI5à^

SAPUI5 is a JavaScript framework for building cross-platform, enterprise-grade web apps with modern, responsive, and rich user interfaces. For more information, see [Developing SAPUI5](#).

HTML5à^

On SAP BTP, you can access and run HTML5 applications in a cloud environment without the need to maintain your own runtime infrastructure. For more information, see [Developing HTML5 Applications in the Cloud Foundry Environment](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/developing-with-sap-cloud-application-programming-model>

Title: *Developing with the SAP Cloud Application Programming Model | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Developing

Applications and ServicesDeveloping with the SAP Cloud Application Programming Model

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Developing with the SAP Cloud Application Programming Model

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It guides developers along a 'golden path' of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

Use Core Data Services (CDS) to build data models and service definitions on a conceptual level. These CDS models are used as inputs for the data, service, and UI layers. They're then translated to native artifacts, for example SQL database schemas, and interpreted to automatically serve requests at runtime.

In summary, CDS is used as a business level data definition source, and to generate the artifacts at the persistence layer. It's used to define visual aspects relating to the data, with those definitions (annotations) defining the UI layer. And it's used to generate the application service layer.

We provide seamless integration with the Cloud Foundry environment on SAP Business Technology Platform. This makes it easier for you to deploy your application and consume platform services.

The programming model is compatible with any development environment, but we recommend using the SAP Business Application Studio.

The following graphic shows the relationship between CAP, SAP Business Technology Platform, platform services, and development tools:

Overview of the SAP Cloud Application Programming Model

Best Practices

To help you create concise and comprehensible models with CDS, we have put together a list of best practices.

Core Data Services (CDS) Language Reference Documentation

CDS is the backbone of the SAP Cloud Application Programming Model (CAP). It provides the means to declaratively capture service definitions and data models, queries, and expressions in plain (JavaScript) object notations. CDS features to parse from a variety of source languages and to compile them into various target languages.

Developing Business Applications Using Java

The CAP Java SDK enables developing SAP Cloud Application Programming Model (CAP) applications in Java. While the SAP Business Application Studio provides excellent support to develop CAP Java applications, you can also develop locally with your tool of choice, for example Eclipse.

Developing Business Applications Using Node.js

Develop a sample business service using Core Data Services (CDS), Node.js, and SQLite, by using the SAP Cloud Application Programming Model (CAP). Develop on your local environment and deploy to the Cloud.

References

Find additional references in the following sections.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enabling-joule>

Title: Enabling Joule | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Formations for Services in SAP BTPEnabling Joule

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Enabling Joule

Useà^

Note

The content in this section is not relevant for China (Shanghai) and Government Cloud (US) regions.

Leveraging its generative AI capabilities, Joule aims to understand your requests, retrieve information, and complete your tasks in a conversational way. You can describe the current task to Joule, who can help automate tasks or take you to places where you can find desired data. By using Joule, you can easily interact with the system and increase your productivity.

To use Joule in SAP systems, you have to integrate these systems with Joule. To do that, you have to include the SAP systems in a formation of type Integration with Joule on the System Landscape Formations page of the SAP BTP cockpit. The integration is between the Joule system and every SAP system in the formation, but not between each and every SAP system.

To see which SAP systems are supported, see Joule Capabilities.

Rulesà^

When creating Integration with Joule formations, keep in mind the following rules:

At most one system of type Joule can be included in one Integration with Joule formation.

A system of type Joule can be included in at most one Integration with Joule formation.

Applicable to systems of type SAP Integrated Business Planning: you can include these systems in a formation of type Integration with Joule only if they are registered using the All Communication Scenarios communication scenario group. See Register an SAP Integrated Business Planning System in a Global Account in SAP BTP.

Applicable to systems of type SAP SuccessFactors: you can include these systems in a formation of type Integration with Joule only after they have been registered. See Register an SAP SuccessFactors System in a Global Account in SAP BTP.

Applicable to systems of type SAP S/4HANA Cloud: you can include these systems in a formation of type

Integration with Joule only if they are registered using the All Communication Scenarios communication scenario group. See Register an SAP S/4HANA Cloud System in a Global Account in SAP BTP.

Note

This documentation refers to SAP S/4HANA Cloud Public Edition. See Introduction to the Universe of SAP S/4HANA Cloud Public Edition.

Prerequisites

You are a global account administrator, or you are a system landscape administrator. See Working with Role Collections.

You have a Joule system and at least one other system of the supported types that fulfill the rules in the Systems page. The Joule system is auto-discovered.

Procedure

The following procedure outlines the steps you need to perform to enable the integration between Joule and SAP systems.

As an alternative to the steps that follow, you can use the Setting Up Joule booster in the SAP BTP cockpit and have the end-to-end scenario set up. See Run the Joule Booster.

Note

We recommend that you use the Setting Up Joule booster.

In the SAP BTP cockpit, in the System Landscape Systems page of the SAP BTP cockpit, browse the already added systems in your customer system landscape or manually add and register any missing systems.

The Joule system you are going to include in this formation must not be part of another formation.

The customer system landscape features systems that are added to the list in one of the following ways:

Auto-discovered

An auto-discovered system is a system (associated with the given global account) that has been discovered and added automatically to the list based on information of the existing system landscape. Any SAP system of the supported system types that is associated with the same customer ID, with which your global account in SAP BTP is associated, will be added automatically in the system landscape list.

Subaccount/<my-subaccount>

A subscription in SAP BTP cockpit associated with a given subaccount. The subscription has been discovered and added automatically through the subaccount. Check the value of the Discovery column to see the subaccount where your system is subscribed.

Manually added

Specifies that the system has been added to the list manually by the global account administrator, using the Add System button and completing the wizard. The system has been associated with the global account in SAP BTP.

Note

If a given SAP system is missing on the Systems page, it may be associated with a different customer ID on the SAP BTP global account you are working in. In this case, you need to add the system manually, and then, register it.

See Adding, Registering and Deregistering Systems.

Create a formation of type Integration with Joule and include the relevant systems in it.

Add any name that helps you identify your formation.

In the Formation Type dropdown menu, select Integration with Joule.

Select the systems that you want to include in the formation. One of these systems must be of type Joule.

Note

Systems can only be added to one formation of type Integration with Joule in a global account.

Also, a formation of type Integration with Joule can contain only one system of type Joule.

Review your selections and create the formation.

Next Steps

The next steps of the different SAP solutions integrated with Joule are listed at Integrating Joule with SAP Solutions.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enabling-system-landscape-for-sap-build>

Title: Setting Up System Landscape for SAP Build | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Formations for ExtensibilitySetting Up System Landscape for SAP Build

The content in this section is not relevant for China (Shanghai) and Government Cloud (US) regions.

A registered SAP S/4HANA Cloud system in the SAP BTP cockpit can expose consumption bundles that contain APIs and events. You can easily discover and consume the APIs exposed by the SAP S/4HANA Cloud systems in your system landscape when you develop and extend applications on SAP BTP, Cloud Foundry environment, using SAP Build. To do this, first you need to enable connectivity between your system landscape in SAP BTP cockpit and SAP Build. The integration requires performing several configuration steps starting with configuration on a global account level, and then, configuration on a subaccount level.

Note

This documentation refers to SAP S/4HANA Cloud Public Edition. See Introduction to the Universe of SAP S/4HANA Cloud Public Edition.

Prerequisitesà^

You are a global account administrator, or you are a system landscape administrator. See Working with Role Collections.

You have an SAP Build subscription.

Procedureà^

The following procedure outlines the steps you need to perform to consume the APIs of registered SAP S/4HANA Cloud systems within SAP Build.

Add and then register an SAP system of type SAP S/4HANA Cloud in the System Landscape Systems page of SAP BTP cockpit.

To expose information about its APIs and events and show this information on the Systems page, an SAP system of type SAP S/4HANA Cloud must be registered in the SAP BTP cockpit. Only when registered, the system communicates information about its APIs and other technical details across the landscape. See Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment.

In the Formations page, create a formation of type Integration with SAP Build and include the SAP Build system and the SAP S/4HANA Cloud systems you want to expose in SAP Build.

Note

In the formation type Integration with SAP Build, you can include only SAP systems of type SAP Build and SAP S/4HANA Cloud.

To enable connectivity between given SAP systems of type SAP S/4HANA Cloud from the Systems page of SAP BTP cockpit and SAP Build, you must create a formation of the corresponding type and include the SAP S/4HANA Cloud systems in it. See Automating Integrations Using Formations.

Create a destination and make sure it has the corresponding system and consumption bundle properties.

Create a destination manually.

You can create the destination manually via the SAP Destination service page in SAP BTP cockpit by specifying one of the following additional properties variants in it.

Additional Properties (variant 1)

Additional Properties (variant 2)

Description

x-correlation-id

x-correlation-id

The property identifies the consumption bundle and its APIs that are exposed by the SAP S/4HANA Cloud system. The destination also provides the required credentials to consume the bundle and use it for further development in SAP Build.

x-system-id

x-system-name

Properties that uniquely identify the local tenant of the registered SAP S/4HANA Cloud system.

x-system-type

x-system-base-url

All of the required properties of a given SAP S/4HANA Cloud system are accessible in the corresponding System Details section on the Systems page.

Create a destination automatically.

Alternatively, the destination can be created automatically. You can do this, by creating a service instance of the SAP S/4HANA Cloud Extensibility service after you register an SAP S/4HANA Cloud system. See [Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment and Create a Service Instance to Consume the SAP S/4HANA Cloud APIs](#).

Next Steps

Develop applications on SAP BTP, Cloud Foundry environment using SAP Build and the system landscape. When you enable the connectivity, the SAP S/4HANA Cloud systems and their APIs are accessible in SAP Build. Next you can use the APIs to develop new or extend the existing functionality with the help of SAP Build.

Related Information

SAP Build Process Automation: Using SAP Systems

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enabling-system-landscape-for-sap-business-application-studio>

Title: *Setting Up System Landscape for SAP Business Application Studio | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Formations for ExtensibilitySetting Up System Landscape for SAP Business Application Studio

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Setting Up System Landscape for SAP Business Application Studio

Use

Note

The content in this section is not relevant for China (Shanghai) and Government Cloud (US) regions. A registered SAP S/4HANA Cloud system in the SAP BTP cockpit can expose consumption bundles that contain APIs and events. You can easily discover and consume the APIs exposed by the SAP S/4HANA Cloud systems in your system landscape when you develop and extend applications on SAP BTP, Cloud Foundry environment, using SAP Business Application Studio. To do this, first you need to enable connectivity between your system landscape in SAP BTP cockpit and SAP Business Application Studio. The integration requires performing several configuration steps starting with configuration on a global account level, and then, configuration on a subaccount level.

Note

This documentation refers to SAP S/4HANA Cloud Public Edition. See [Introduction to the Universe of SAP S/4HANA Cloud Public Edition](#).

Prerequisites

You are a global account administrator, or you are a system landscape administrator. See [Working with Role Collections](#).

You have an SAP Business Application Studio subscription. See [Subscribe to SAP Business Application Studio](#).

Procedure

The following procedure outlines the steps you need to perform to consume the APIs of registered SAP S/4HANA Cloud systems within the SAP Business Application Studio.

In the System Landscape Systems page of SAP BTP cockpit, add and then register an SAP system of type SAP S/4HANA Cloud.

To expose information about its APIs and events and show this information on the Systems page, an SAP system of type SAP S/4HANA Cloud must be registered in the SAP BTP cockpit. Only when registered, the system communicates information about its APIs and other technical details across the landscape. See [Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment](#).

In the System Landscape Formations page, create a formation of type Developing with SAP Business Application Studio and include the SAP Business Application Studio system and the SAP S/4HANA Cloud systems you want to expose in SAP Business Application Studio.

Note

In the formation type Developing with SAP Business Application Studio, you can include only SAP systems of type SAP Business Application Studio and SAP S/4HANA Cloud.

To enable connectivity between given SAP systems of type SAP S/4HANA Cloud from the Systems page of SAP BTP cockpit and the SAP Business Application Studio, you must create a formation of the corresponding type and include the SAP S/4HANA Cloud systems in it. See Automating Integrations Using Formations. Create a destination and make sure it has the corresponding system and consumption bundle properties. Create a destination manually.

You can create the destination manually via the SAP Destination service page in SAP BTP cockpit by specifying one of the following additional properties variants in it.

Additional Properties (variant 1)

Additional Properties (variant 2)

Description

x-correlation-id

x-correlation-id

The property identifies the consumption bundle and its APIs that are exposed by the SAP S/4HANA Cloud system. The destination also provides the required credentials to consume the bundle and use it for further development in the SAP Business Application Studio.

x-system-id

x-system-name

Properties that uniquely identify the local tenant of the registered SAP S/4HANA Cloud system.

x-system-type

x-system-base-url

All of the required properties of a given SAP S/4HANA Cloud system are accessible in the corresponding System Details section on the Systems page.

Create a destination automatically.

Alternatively, the destination can be created automatically. You can do this, by creating a service instance of the SAP S/4HANA Cloud Extensibility service after you register an SAP S/4HANA Cloud system. See Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment and Create a Service Instance to Consume the SAP S/4HANA Cloud APIs.

Next Steps

Develop applications on SAP BTP, Cloud Foundry environment using SAP Business Application Studio and the system landscape.

When you enable the connectivity, the SAP S/4HANA Cloud systems and their APIs are accessible in the SAP Business Application Studio. Next you can use the APIs to develop new or extend the existing functionality with the help of SAP Business Application Studio. See Unified Customer Landscape Service Provider.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/environments>

Title: Environments | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsEnvironments

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Environments

Environments constitute the actual platform-as-a-service offering of SAP BTP that allows for the development and administration of business applications. Environments are anchored in SAP BTP on subaccount level.

Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. So a multi-environment subaccount is your single address to host a variety of applications and offer diverse development options. One advantage of using different environments in one subaccount is that you only need to manage users, authorizations, and entitlements once per subaccount, and thus grant more flexibility to your developers.

Note

The multi-environment subaccount functionality is not applicable for the Neo environment.

Environment Instances

To actually use an environment in a subaccount, you must enable it, which creates an instance of that environment. There are several ways to create environment instances:

In the SAP BTP cockpit, on the subaccount overview, choose Enable.

In the SAP BTP cockpit, under Service Marketplace. Here, you get more information, such as the available plans and links to further information.

Using the `btcp CLI` command `btcp create accounts/environment-instance`

Cloud Foundry Environment

The Cloud Foundry environment allows you to create polyglot cloud applications in Cloud Foundry. It contains the SAP BTP, Cloud Foundry runtime service, which is based on the open-source application platform managed by the Cloud Foundry Foundation.

ABAP Environment

The ABAP environment allows you to create extensions for ABAP-based products, such as SAP S/4HANA Cloud, and develop new cloud applications. You can transform existing ABAP-based custom code or extensions to the cloud.

Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

Related Information

Account Administration

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/html5-application-repository>

Title: HTML5 Application Repository | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Developing HTML5 Applications in the Cloud Foundry EnvironmentHTML5 Application Repository

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

HTML5 Application Repository

HTML5 application repository enables central storage of HTML5 applications' static content on the SAP BTP, Cloud Foundry environment. This service can be consumed from the SAP BTP, Cloud Foundry runtime, the SAP BTP, Kyma runtime, Kubernetes, or by creating a service instance on the subaccount level regardless of an environment.

HTML5 applications consist of static content such as HTML, CSS, JavaScript, and other files, that run on a browser. For more information, see Basic Template and `openui5-basic-template-app`.

HTML5 application repository allows application developers to manage the lifecycle of their HTML5 applications. In runtime, the repository enables the consuming application, typically the application router, to access HTML5 application static content in a secure and efficient manner. For more information, see Configure Application Router.

Features

Zero Down-Time Enablement

The HTML5 applications are decoupled from the consuming application router. This enables updating the static content of the HTML5 applications without restarting the application router.

Versioning and Authorization

Exploration of HTML5 application content by version.

Access control that is based on private or public authorization.

When the HTML5 application is public, the service enables sharing this content with consuming application routers from different spaces.

Availability and Performance

During runtime, the HTML5 application content is cached and optimized to provide high performance with minimal network load.

The service provides several instances for a runtime to serve a high load of application requests.

Environment

This service is available in the following environments:

Cloud Foundry environment

Kyma environment

Multitenancy Support

This service supports multitenancy. It can be used by tenant-aware applications.

Technical Constraints

The size of an application deployed to the repository is limited to 100 MB per service instance of the app-host service plan.

Since the applications stored in HTML5 application repository can be shared, it is advised not to add personal data to them.

You can perform up to 50 deployments per minute at design time per SAP BTP subaccount.

If you exceed this rate limit, the HTTP error code 429 is returned.

You can make up to 300 requests per second at runtime per SAP BTP subaccount.

If you exceed this rate limit, the HTTP error code 429 is returned.

See Also

If you want to learn more about services in the SAP BTP, Cloud Foundry environment, see [Using Services in the Cloud Foundry Environment](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/managed-application-router>

Title: [Managed Application Router | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Developing HTML5 Applications in the Cloud Foundry EnvironmentManaged Application Router

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Managed Application Router

The managed application router enables you to access and run HTML5 applications in a cloud environment without the need to maintain your own runtime infrastructure.

The managed application router is the HTML5 applications runtime capability that is provided by the following products:

SAP Build Work Zone, standard edition

SAP Build Work Zone, advanced edition

SAP Cloud Portal

To use the managed application router, you must be subscribed to one of these services.

For information how to develop HTML5 applications and run them using the managed application router, see the following product-specific guides:

[Developing HTML5 Applications for SAP Build Work Zone, standard edition](#)

[Developing HTML5 Applications for SAP Build Work Zone, advanced edition](#)

[Developing HTML5 Applications for SAP Cloud Portal](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/neo-environment>

Title: [Neo Environment | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform ConceptsEnvironmentsNeo Environment

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

See also:

[What Is SAP BTP, Neo Environment](#)

[Development, Neo Environment](#)

[Extensions, Neo Environment](#)

[Administration and Operations, Neo Environment](#)

Security, Neo Environment

Getting Support, Neo Environment

Migration from the Neo to the Multi-Cloud Foundation

To learn more about why and how to migrate your scenarios on SAP BTP, see [Migrating from the SAP BTP Neo Environment to the Multi-Cloud Foundation](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/platform-users-440131674b0147a09e696053a53832d5>

Title: Platform Users | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsUser and Member ManagementPlatform Users

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

Platform Users

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot accounts, applications and services on SAP BTP. They also manage entitlements and environments. They're the users that have full access and give certain permissions, for instance, at global account, directory, or subaccount level. Due to their powerful authorizations, we recommend that you protect them with strong authentication (for example, two-factor authentication). Members of subaccounts only have basic access. Platform users who have administrative permissions can view or manage the list of global accounts, subaccounts, and environments, such as Cloud Foundry orgs and spaces. Platform users have access to them using the SAP BTP cockpit, infrastructure as code, the SAP BTP command-line interface (btp CLI), or environment-specific CLI, such as the Cloud Foundry (CF) CLI. Using Joule is bound to the SAP BTP cockpit authorization.

For platform users, there's a default identity provider. We expect that you have your own identity provider.

Recommendation

We strongly recommend that you configure your custom tenant of SAP Cloud Identity Services as the identity provider and connect SAP Cloud Identity Services to your own corporate identity provider.

For more information, see [Trust and Federation with Identity Providers](#).

Note

For China (Shanghai) and Government Cloud (US) regions, a different default identity provider is used, and you can't use SAP Cloud Identity Services as identity provider in the global account.

If you want to use two-factor authentication in the China (Shanghai) region, see [this blog article on SAP Community](#).

Member Management

Member management refers to managing permissions for platform users. Members have only basic access to SAP BTP.

Member management happens at global account, directory, subaccount, and environment level. Members' permissions apply to all operations that are associated with the global account, the organization, or the space, irrespective of the tool used. Depending on the scope and the cloud management tools you're using, you manage members in different ways:

Global Accounts

Directories

Subaccounts

You manage global account members by assigning role collections to platform users. Use the following predefined role collections:

Global Account Administrator

Global Account Viewer

Assign these role collections from the SAP BTP cockpit or the btp CLI.

See:

[Role Collections and Roles in Global Accounts, Directories, and Subaccounts](#)

[Add Members to Your Global Account](#)

[Create Users](#)

You manage directory members by assigning role collections to platform users. Use the following predefined

role collections:

Directory Administrator

Directory Viewer

Assign these role collections from the SAP BTP cockpit or the btp CLI.

See:

Role Collections and Roles in Global Accounts, Directories, and Subaccounts

Create Users

Manage Users in Directories

You manage subaccount members by assigning role collections to platform users.

Note

Neo subaccounts don't use role collections.

For more information, see Managing Member Authorizations in the Neo Environment.

Use the predefined role collections, such as:

Subaccount Administrator

Subaccount Viewer

Assign these role collections from the SAP BTP cockpit or the btp CLI.

See:

Role Collections and Roles in Global Accounts, Directories, and Subaccounts

Add Members to Your Subaccount

Create Users

Member Management in the Cloud Foundry Environment

Orgs

Spaces

Manage org members on the Org Members page at environment level in the subaccount using the SAP BTP cockpit or with the Cloud Foundry CLI.

A platform user added as an org member can be either an Org Manager or an Org Auditor or implicitly an Org User.

See:

Org Administration Using the Cockpit

About Roles in the Cloud Foundry Environment

<https://docs.cloudfoundry.org/concepts/roles.html#roles>

Add Org Members

Manage space members on the Org Members page at space level in the SAP BTP cockpit or with the Cloud Foundry CLI.

A platform user added as a space member can be either a Space Manager, Space Developer, Space Auditor, or Space Supporter.

See:

About Roles in the Cloud Foundry Environment

<https://docs.cloudfoundry.org/concepts/roles.html#roles>

Add Space Members

à

For more information on the Kyma environment, see Assign Roles in the Kyma Environment.

See also About User Management in the Cloud Foundry Environment.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/sap-business-technology-platform>

Title: SAP Business Technology Platform | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform

SAP Business Technology PlatformEnglish

Provide feedback on our search

à

SAP Business Technology Platform

SAP Business Technology Platform (SAP BTP) is an offering that integrates the following technology portfolios: application development; process automation; integration; data, analytics, and enterprise planning; artificial intelligence. The platform allows users to turn data into business value, compose end-to-end business processes, connect entire IT landscapes, and personalize, build, and extend SAP applications. This reduces

the overall total cost of ownership by maintaining SAP landscapes and third-party software across end-to-end business processes.

SAP BTP in the Intelligent Enterprise[^]

Companies need access to live data to make informed decisions in real time. They also need to apply advanced technologies and best practices within agile, integrated business processes. That's why a key part of SAP's strategy is integrating end-to-end processes, whether the solutions come from SAP, partners, or third parties.

SAP drives integration across its solution portfolio in the following business scenarios: lead to cash, source to pay, design to operate, and hire to retire.

SAP Business Technology Platform offers integration capabilities that connect end-to-end business processes across SAP and third-party applications. The following key aspects, known as Suite Qualities, provide a consistent experience across applications:

Seamless User Experience: SAP Fiori gives many SAP solutions a common look and feel. This improves user experience and reduces interruptions.

See: SAP Fiori.

Consistent Security and Identity Management: SAP Cloud Identity services on SAP BTP enable you to manage identities and use the single sign-on capability across selected end-to-end processes.

See: SAP Cloud Identity Services on SAP BTP.

Aligned Domain Models, APIs, and Events: Master Data Integration is a cloud service for master data integration. It provides a consistent view of master data across a hybrid landscape. You can use APIs for process integration among SAP solutions, SAP, and third parties, or by adopting an event-based integration strategy – all based on predefined integration content available on SAP API Business Hub.

See: SAP One Domain Model, SAP Master Data Integration service, SAP Integration Suite on SAP BTP

Embedded Analytics across Solutions: The analytical insights in various SAP solutions help you make informed decisions. Embedded analytics from SAP Analytics Cloud are available, for example, in SAP S/4HANA and SAP SuccessFactors.

See: SAP Analytics Cloud on SAP BTP.

One Workflow Inbox: With the unified view of pending tasks across SAP solutions in both mobile and desktop environments, you can complete tasks faster.

See: SAP Task Center on SAP BTP, SAP Mobile Start. For full workflow modeling and visibility capabilities leverage, see: SAP Process Automation on SAP BTP

Coordinated Lifecycle Management: You can optimize implementation projects and reduce configuration effort because provisioning, setup and operations, and monitoring solutions are harmonized. Automated provisioning and guided integration setup are available for selected scenarios in lead to cash, source to pay, recruit to retire, and design to operate.

See: SAP for Me, SAP CALM

End-to-End Process Blueprints: Process blueprints that follow the Industry Reference Architecture standard can reduce planning effort for implementation projects. They also support architecture decisions.

See: SAP API Business Hub

Related information:

SAP's vision for the Intelligent Enterprise

SAP's Integration Strategy whitepaper

SAP Integration Strategy community

About This Guide[^]

Get an overview of the basic platform concepts of SAP BTP and how they relate to each other. Learn how to manage your accounts and how to develop, extend, administer, and secure your cloud setup and applications on the platform.

Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

Feedback Create issue: Provide feedback about a documentation page. This option opens an issue on GitHub.

Feedback Edit page: Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

[Contribution Guidelines](#)

[Introduction Video](#)

[Introduction Blog Post](#)

More resources that you might find interesting:

[Best Practices](#)

Get onboarded to SAP BTP. Learn about basic concepts and see our best practices for your development projects in setting up the correct organizational structure, creating an account and security model, and developing and operating applications: [SAP BTP Administrator's Guide](#).

[Services and Solutions](#)

Get an overview on the availability of SAP BTP solutions and services according to region, infrastructure provider, and license model. Access the service details pages for information on features, service plans, related tools, APIs, and documentation: [SAP Discovery Center: Service Catalog](#).

[Neo Environment](#)

For more information about the Neo environment, see [SAP BTP, Neo Environment](#).

To learn more about why and how to migrate your scenarios in the Neo environment, see [Migrating from the Neo Environment to the Multi-Cloud Foundation](#).

[All SAP BTP Resources](#)

You can find all of our resources on the product page: [SAP Business Technology Platform \(SAP BTP\)](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/solutions-and-services>

Title: [Solutions and Services](#) | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsSolutions and Services

SAP Business Technology PlatformEnglish

[Provide feedback on our search](#)

à

Solutions and Services

Consume the solutions and services by SAP BTP according to your preferred development environment and use cases.

[Solutions](#)

SAP BTP offers fast in-memory processing, sustainable, agile solutions and services to integrate data and extend applications, and fully embedded analytics and intelligent technologies.

[Services](#)

Services enable, facilitate, or accelerate the development of business applications and other platform services on SAP BTP. Services are grouped into the following service types:

Business services: Services that enable, facilitate, or accelerate the development of business process components or provide industry-specific functionalities or content within a business application.

Technical services: Services that enable, facilitate, or accelerate the development of general or domain independent content within a business application, independent of the application's business process or task.

You find all available services, solutions, and use cases in the [SAP Discovery Center](#).

Use the service catalog to access service-specific resources:

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/tools>

Title: [Tools](#) | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsTools, Programming Models, Programming Languages, and APISTools

SAP Business Technology PlatformEnglish

[Provide feedback on our search](#)

à

Tools

SAP BTP includes many tools to help you develop and manage applications, and connect them to your on-premise systems.

For more information, see [Tools](#).

Tool

Description

Account Administration in the Cockpit

The SAP BTP cockpit is the web-based administration interface of SAP BTP and provides access to a number of functions for configuring and managing applications, services, and subaccounts. Use the cockpit to manage resources, services, security, monitor application metrics, and perform actions on cloud applications.

Used for account administration in SAP BTP.

SAP Business Application Studio

Designed and optimized for business application development in SAP ecosystems, SAP Business Application Studio enhances productivity by offering specialized tools for various scenarios, including SAP Fiori application development, SAP HANA native extensions, full-stack and mobile application development, and more.

Central to the development environment is Code-OSS, the open-source foundation of Visual Studio Code, ensuring a familiar experience for developers when creating SAP-centric applications. SAP Business Application Studio streamlines the building, testing, and deployment of applications with integrated features for source control and testing. Furthermore, its Full-Stack Application Productivity Toolkit offers intuitive visual tools covering the entire development process, guaranteeing seamless integration with various SAP services and solutions.

Used for:

SAP Fiori development for ABAP and non-ABAP

CAP full-stack application development

SAP Build

SAP Build enables everyone, no matter the skill level, to create and augment enterprise applications, process automations, and business sites with drag-and-drop simplicity.

Used for low-code, no-code development.

Command Line Interface for Cloud Foundry

The Cloud Foundry command line interface enables you to work with the Cloud Foundry runtime to deploy and manage your applications.

Used for managing subaccounts in the SAP BTP, Cloud Foundry runtime.

SAP BTP Command Line Interface

The SAP BTP command line interface (btp CLI) is the command line tool for convenient account management, such as managing global accounts, directories, subaccounts, entitlements, environment instances, multitenant application subscriptions, and users and their authorizations.

Used for managing global accounts in SAP BTP.

Cloud Connector

The Cloud Connector serves as the link between on-demand applications in SAP BTP. This is the browser-based and existing on-premise systems. You can control the resources available for the cloud applications in those systems.

Used for connectivity between SAP BTP applications and on-premise systems.

Terraform provider for SAP BTP

The Terraform provider for SAP BTP enables you to automate the provisioning, management, and configuration of resources on SAP BTP.

kubectI

The Kubernetes command line tool to communicate with a Kubernetes cluster's control plane, using the Kubernetes API.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

kubelogin

A kubectI plugin for Kubernetes OpenID Connect (OIDC) authentication.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Helm

The package manager for Kubernetes, used for installing and managing Kubernetes applications in form of Helm charts.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Paketo (Pack)

Pack is a tool maintained by the Cloud Native Buildpacks project to support the use of buildpacks. Pack lets you

build container images, which are collaboratively maintained making it easier to maintain and update.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

Docker Desktop

Docker Desktop is an application that enables you to manage (build, push, pull, and run) container images on your desktop and a docker-compatible command line interface.

Used as a Kubernetes tool connected with the SAP BTP, Kyma runtime.

SAP Cloud SDK

SAP Cloud SDK provides a layer of abstractions for features of SAP BTP such as logging, multitenancy, and connectivity. It also includes project templates for different execution environments and implementations.

Used for programming in Java and JavaScript.

Eclipse Tool for the Cloud Foundry Runtime

The Eclipse plug-in for the Cloud Foundry runtime is a Java-based toolkit for Eclipse IDE that enables you to develop and deploy Java and Spring applications in the Cloud Foundry runtime from Eclipse or Spring Tool Suite, as well as perform operations such as logging, managing user roles, creating connectivity destinations, and so on.

Used for programming in Java in the SAP BTP, Cloud Foundry runtime.

Service-Specific Tools

The services that run on SAP BTP can come with service-specific tools. For an overview of the services and their tools, see the SAP Discovery Center.
